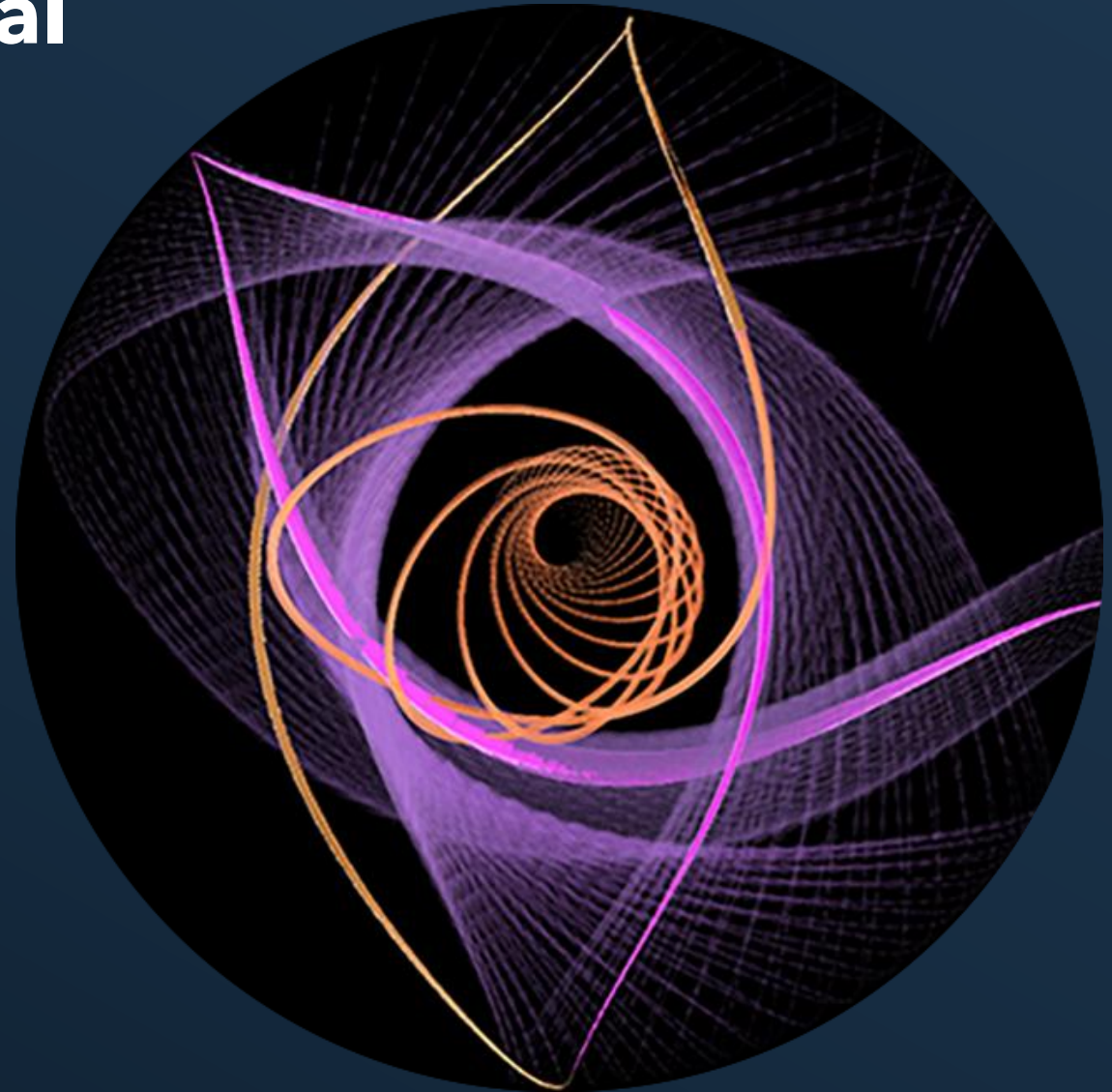


Periodic Orbits and Dynamical Structures in the Earth-Moon Circular Restricted 3-Body Problem

Jannik Gräbner, Jack Sydney Schwartz,
Claudio Vestini

MAE 541 Project

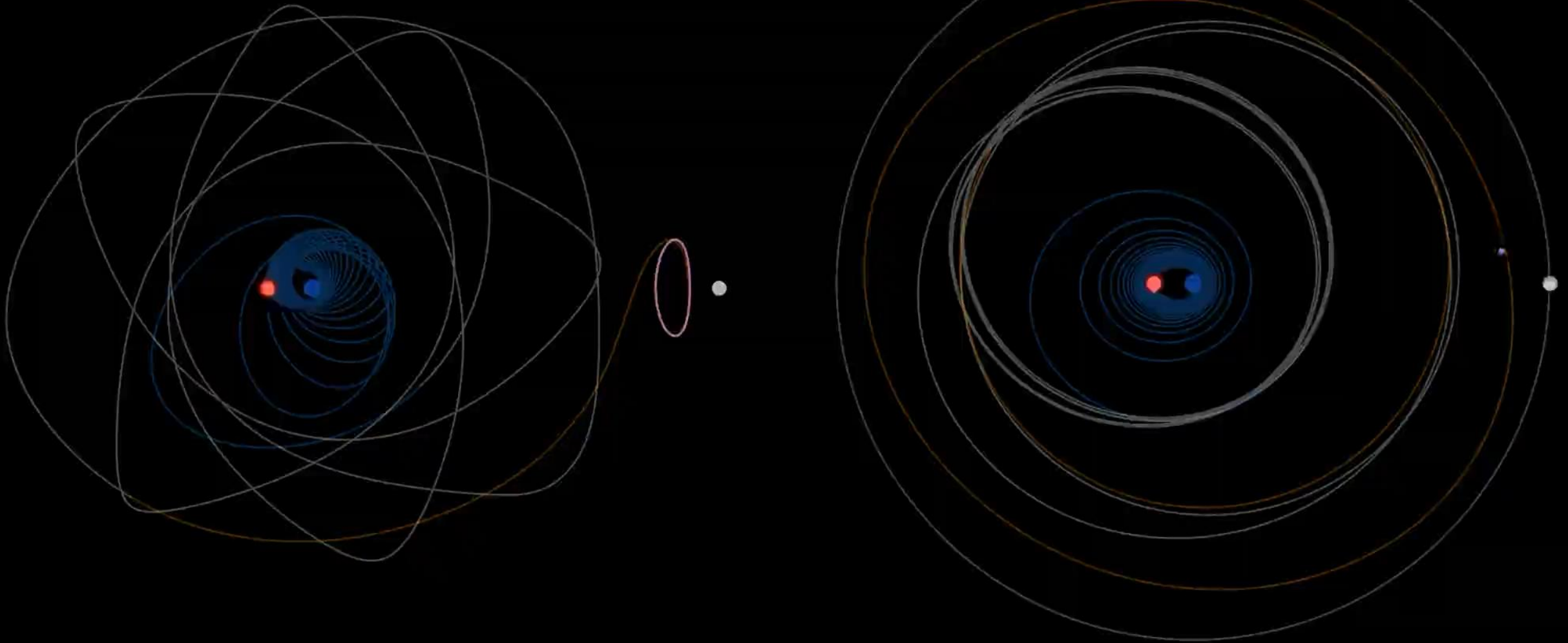


A Spacecraft flying from the Earth to the Moon

Synodic frame

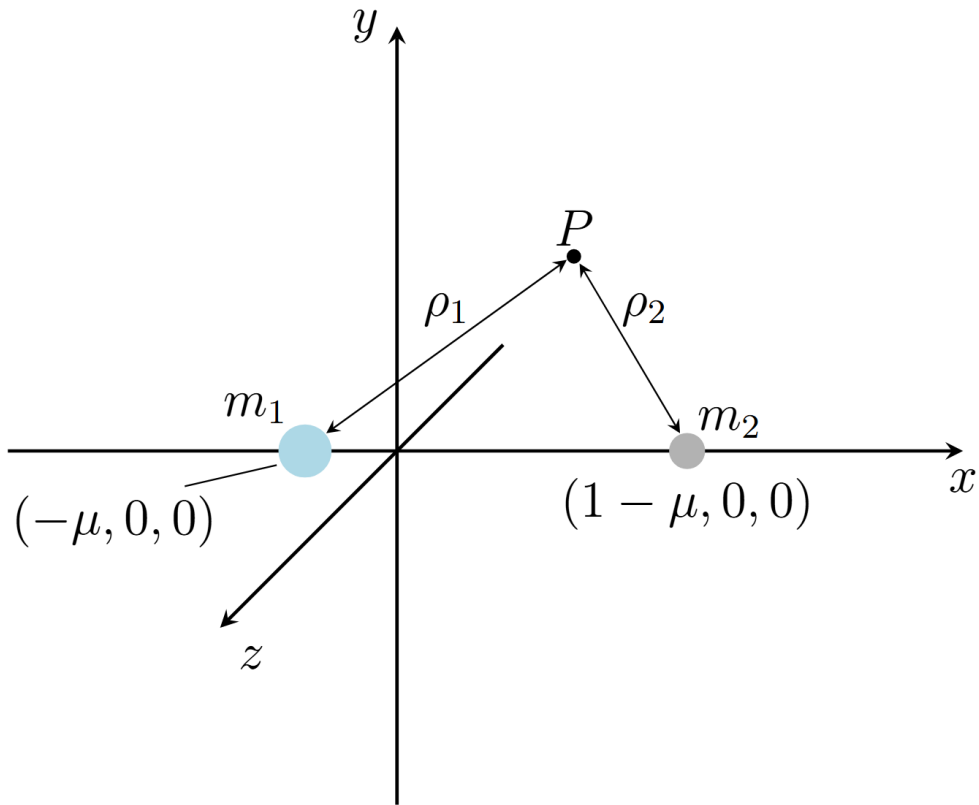
Inertial frame

0.00 days



Circular-Restricted 3-Body Problem (CR3BP)

Synodic Frame:



➤ **Dynamics:**

$$\ddot{\mathbf{r}} = \mathbf{g}(\mathbf{r}, \mathbf{v}) = \begin{pmatrix} 2v_y + x - (1 - \mu)\frac{x+\mu}{\rho_1^3} - \mu\frac{x-1+\mu}{\rho_2^3} \\ -2v_x + y - (1 - \mu)\frac{y}{\rho_1^3} - \mu\frac{y}{\rho_2^3} \\ -(1 - \mu)\frac{z}{\rho_1^3} - \mu\frac{z}{\rho_2^3} \end{pmatrix}$$

$$\rho_1 = \sqrt{(x + \mu)^2 + y^2 + z^2}$$

$$\rho_2 = \sqrt{(x - 1 + \mu)^2 + y^2 + z^2}$$

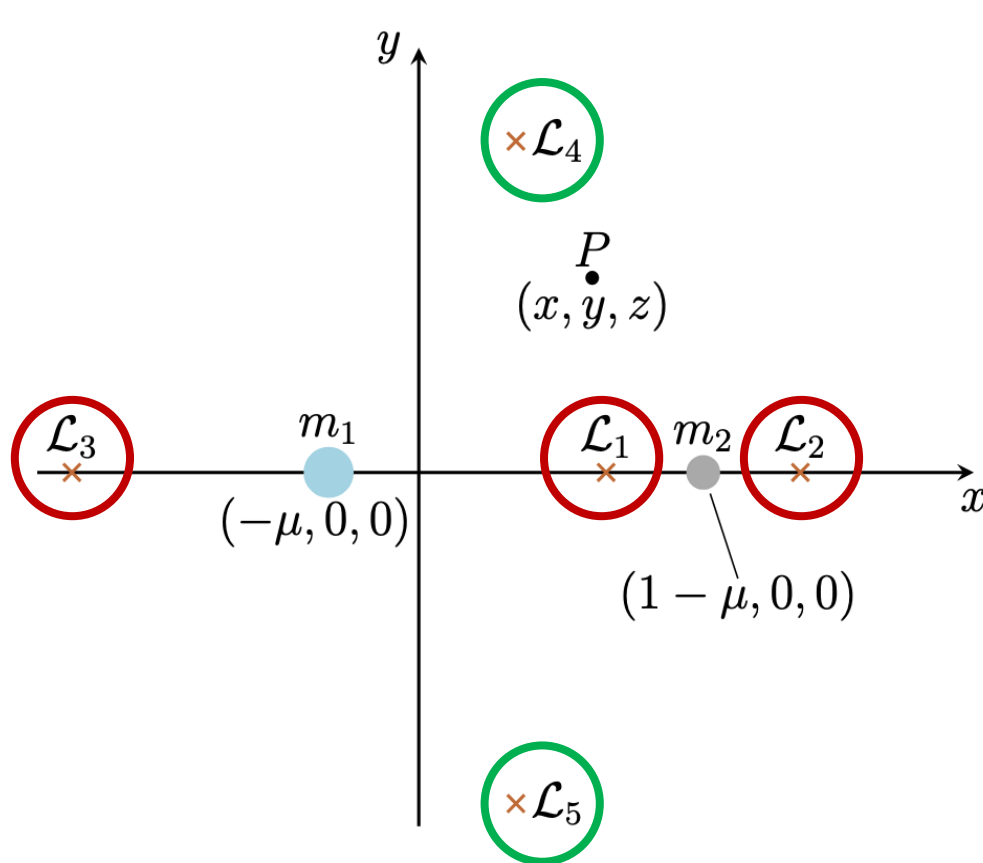
➤ **Mass parameter:** $\mu = \frac{m_2}{m_1 + m_2}$

➤ **Natural units** for mass, time, and distance

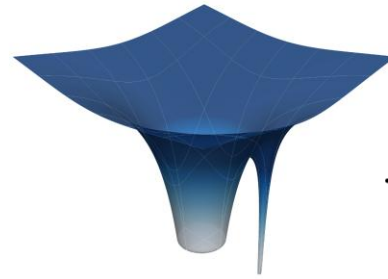
Hamiltonian, Equilibria and Stability

- The system is **Hamiltonian**:

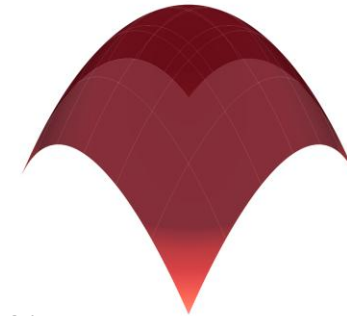
$$H = \tilde{U} + \frac{1}{2} \|\mathbf{v}\|_2^2 = - \left(\frac{1-\mu}{\rho_1} + \frac{\mu}{\rho_2} \right) - \frac{1}{2} (x^2 + y^2) + \frac{1}{2} (v_x^2 + v_y^2 + v_z^2)$$



Gravitational Potential



Centrifugal Potential



Effective Potential



- Equilibrium** points:

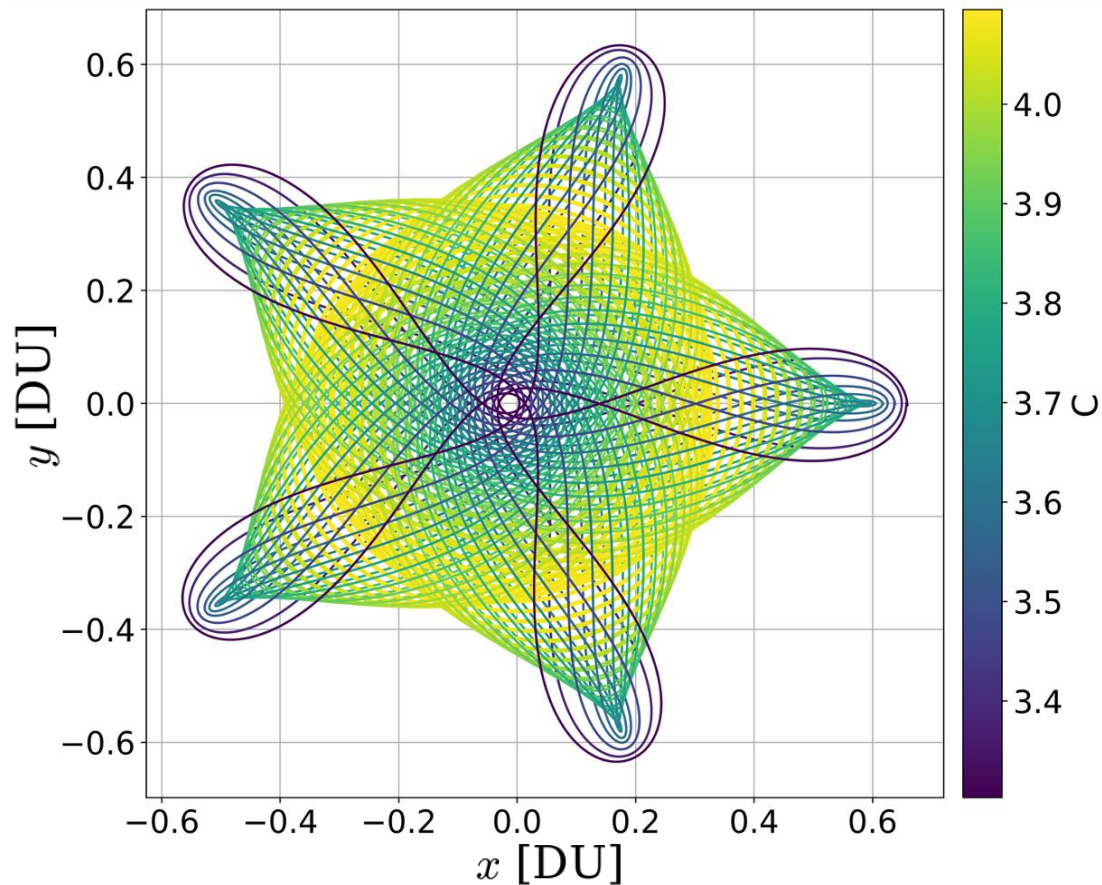
$$\begin{aligned} \mathcal{L}_1 &\approx (0.8369, 0, 0), & \mathcal{L}_4 &= \left(\frac{1}{2} - \mu, \frac{\sqrt{3}}{2}, 0 \right), \\ \mathcal{L}_2 &\approx (1.1557, 0, 0), & \mathcal{L}_5 &= \left(\frac{1}{2} - \mu, -\frac{\sqrt{3}}{2}, 0 \right), \\ \mathcal{L}_3 &\approx (-1.0051, 0, 0). \end{aligned}$$

- Triangular points are **stable** for $\mu < \mu_R$
- Bifurcation** (Klein collision) at $\mu = \mu_R \approx 0.0385$
- This occurs for the **Pluto-Charon** system

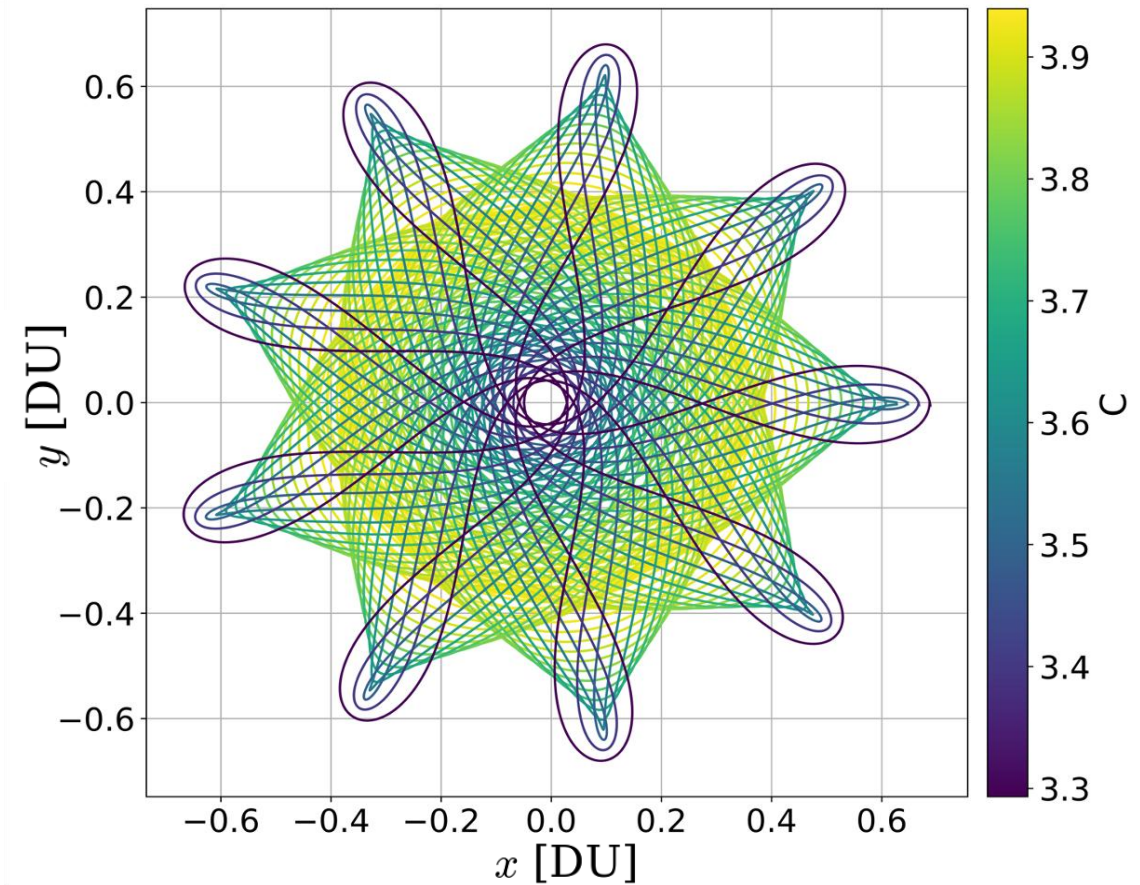
Resonant Orbits in the CR3BP

- Periodic orbits where spacecraft completes p revolutions while primaries complete q ($p:q$)

Prograde 5:1 Family



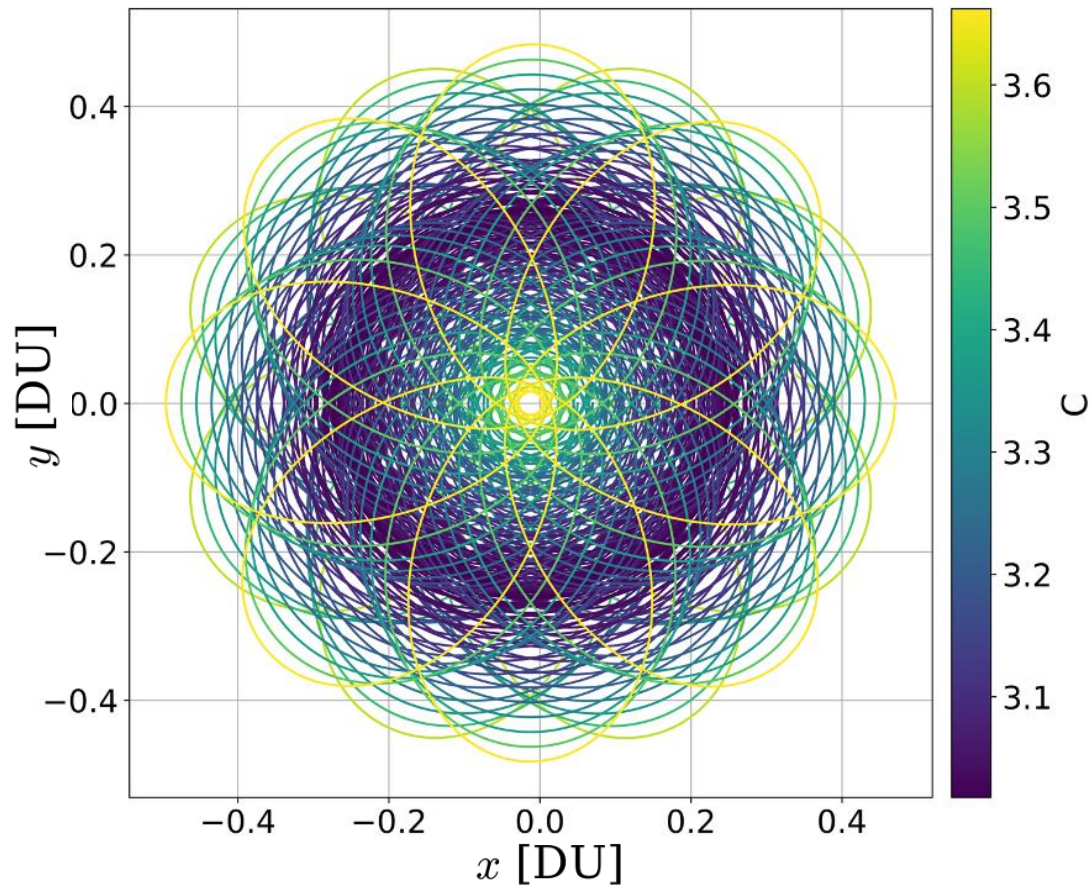
Prograde 9:2 Family



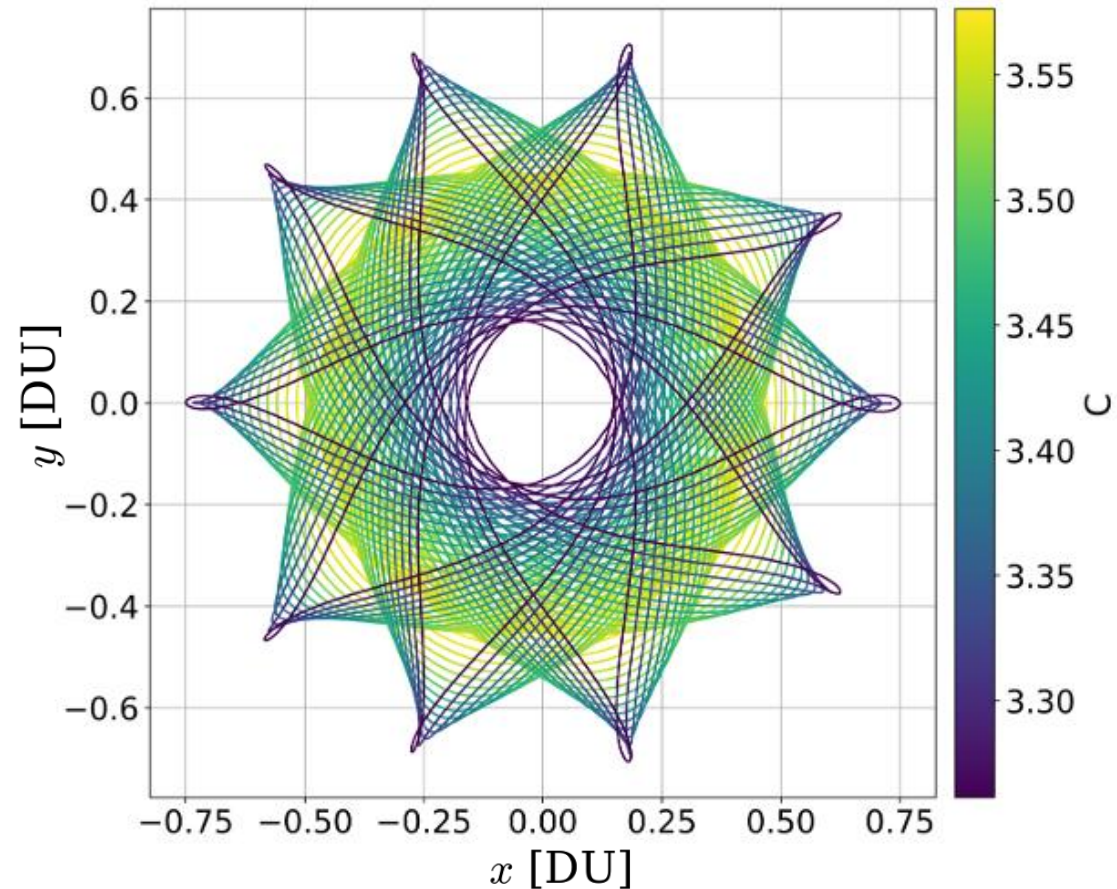
Resonant Orbits in the CR3BP

- Periodic orbits where spacecraft completes p revolutions while primaries complete q ($p:q$)

Retrograde 8:1 Family



Retrograde 10:3 Family



Computation of Periodic Orbits

- Periodic Orbits are computed using **Differential Correction**

Algorithm:

Free variables:

$$\mathbf{X} = \begin{pmatrix} v_{y,0} \\ T/2 \end{pmatrix}$$

Constraints:

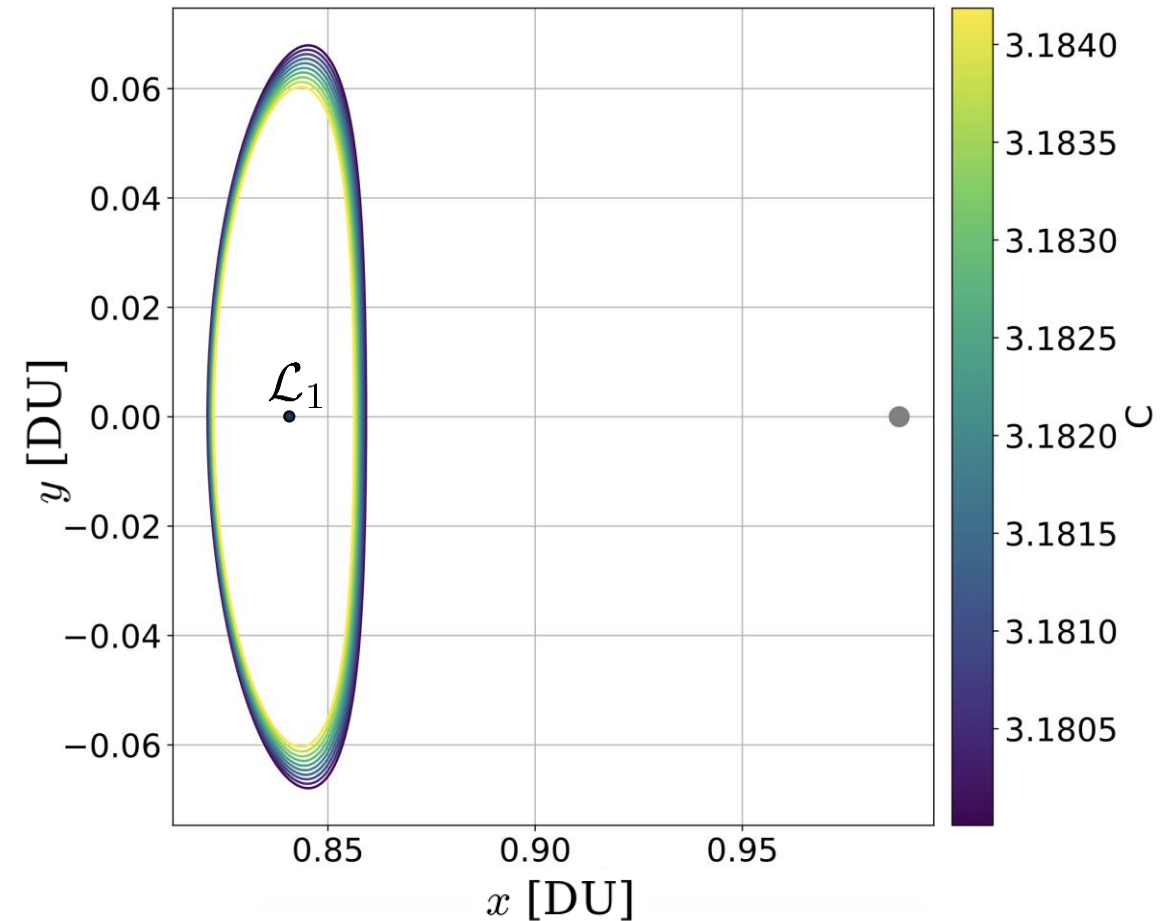
$$\mathbf{F}(\mathbf{X}) = \begin{pmatrix} y(T/2) \\ v_x(T/2) \end{pmatrix} = \mathbf{0}$$

Iterate: $\mathbf{X}^{(k+1)} = \mathbf{X}^{(k)} + \delta \mathbf{X}$

By solving: $\boxed{D\mathbf{F}(\mathbf{X}^{(k)})} \delta \mathbf{X} = -\mathbf{F}(\mathbf{X}^{(k)})$
from STM

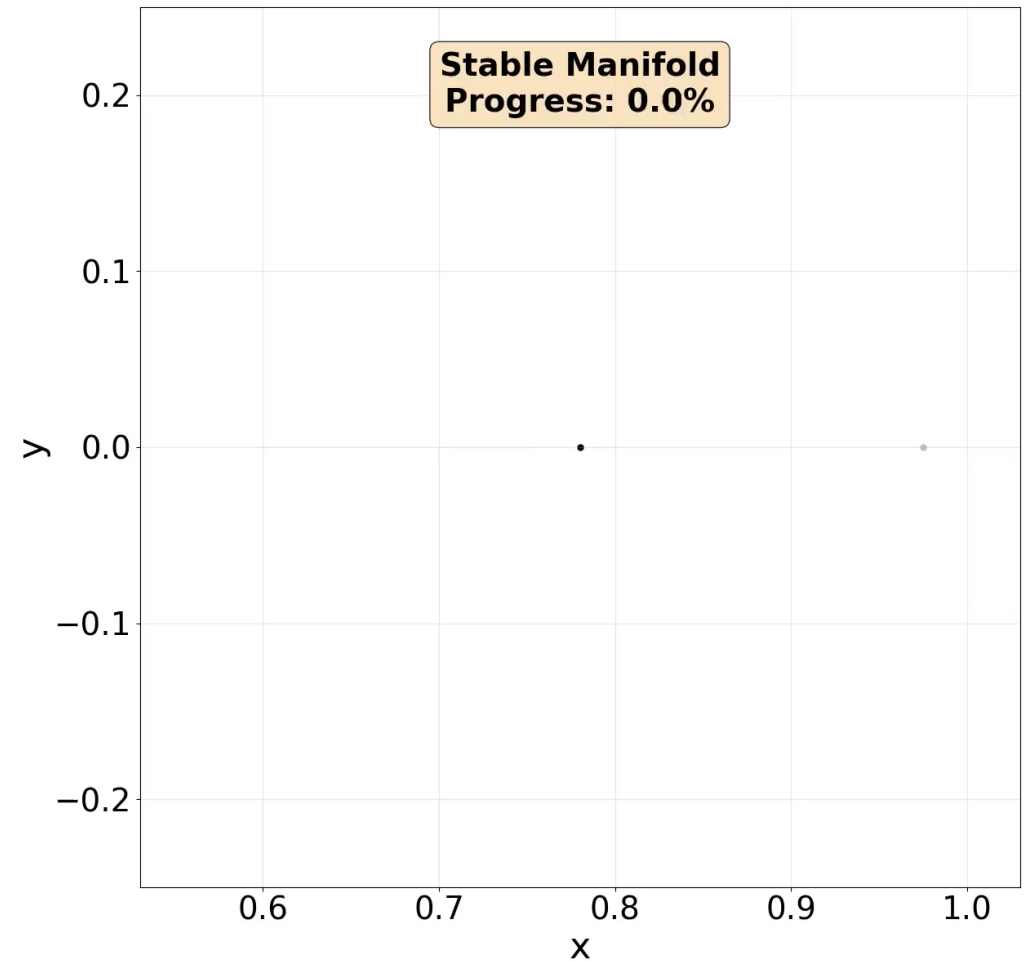
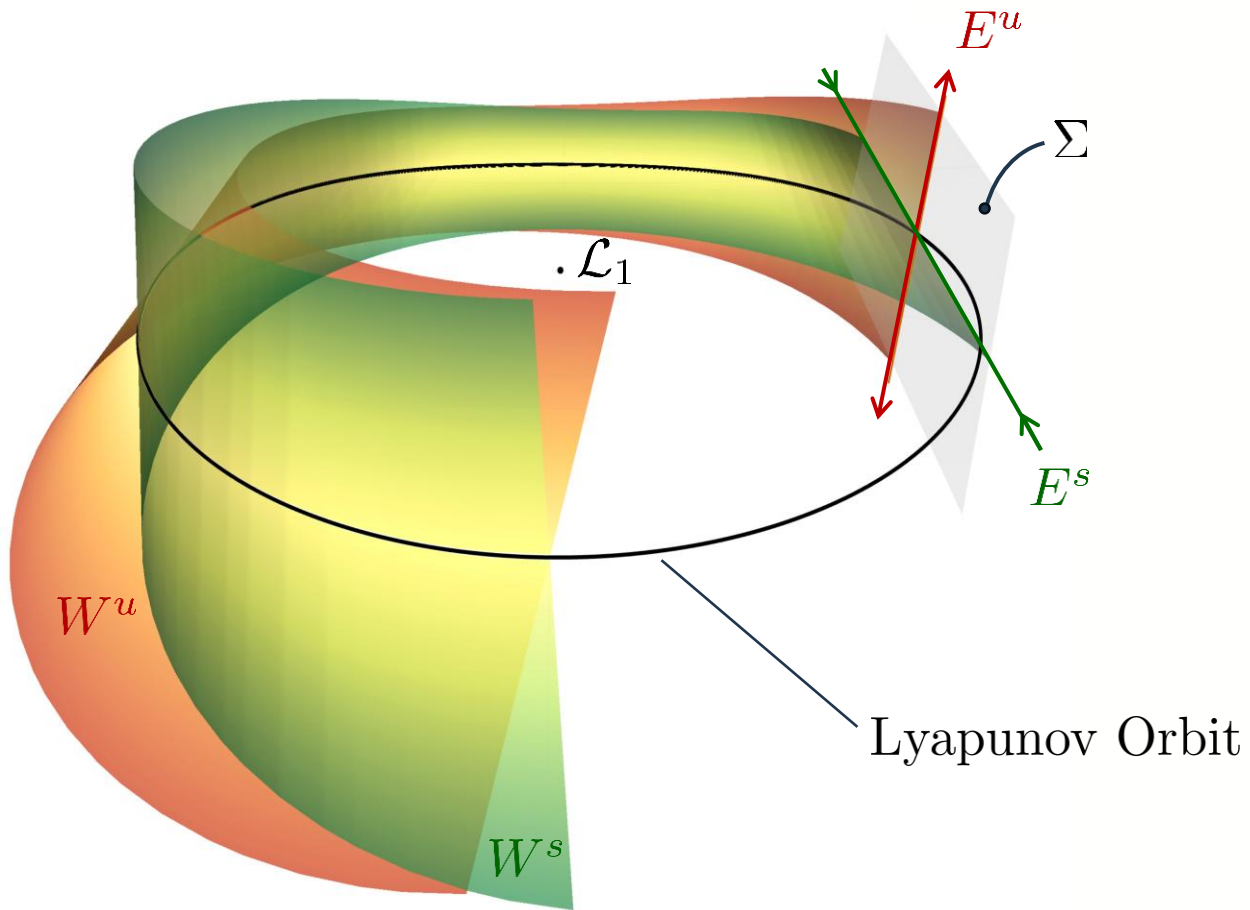
Termination when: $\|\mathbf{F}(\mathbf{X}^{(k)})\| < \varepsilon$

Lyapunov Orbits around $\mathcal{L}_1$



Invariant Manifolds

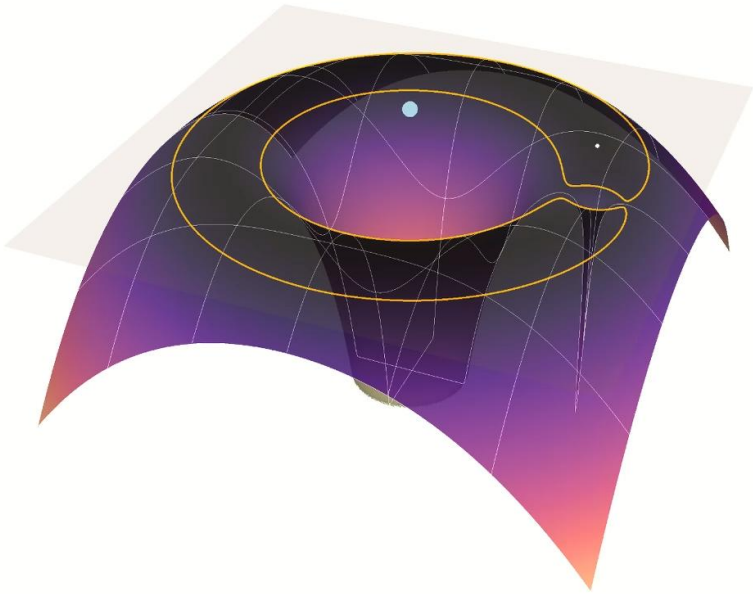
- **Perturb** fixed-energy, unstable periodic orbits along stable and unstable **invariant subspaces**:



Hill's Region and Poincaré Section

- Reachable states are bounded by **Hill's Region** :

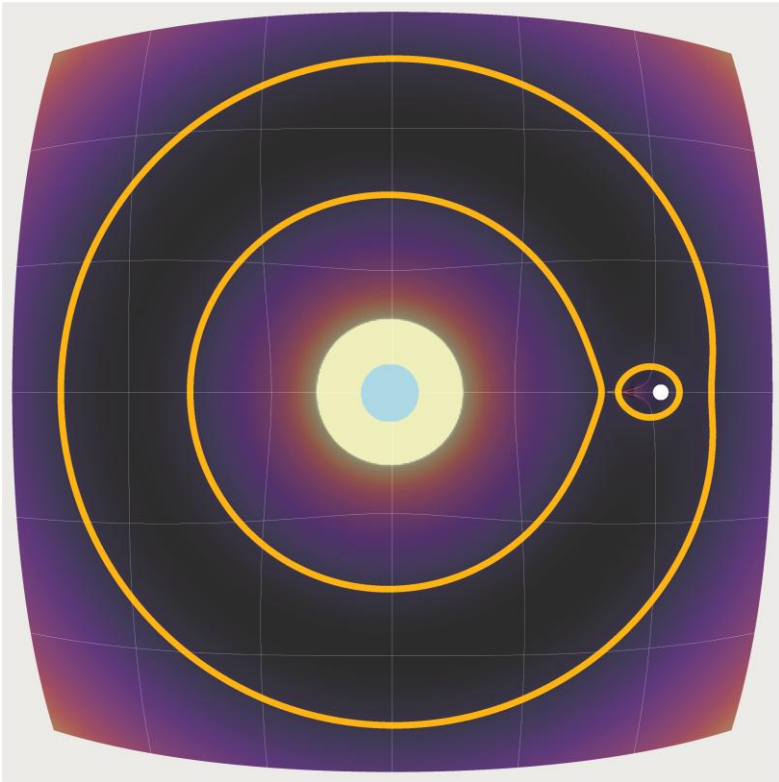
$$\{(x, y, z) : \tilde{U}(x, y, z) \geq C/2\}$$



Hill's Region and Poincaré Section

- Reachable states are bounded by **Hill's Region** :

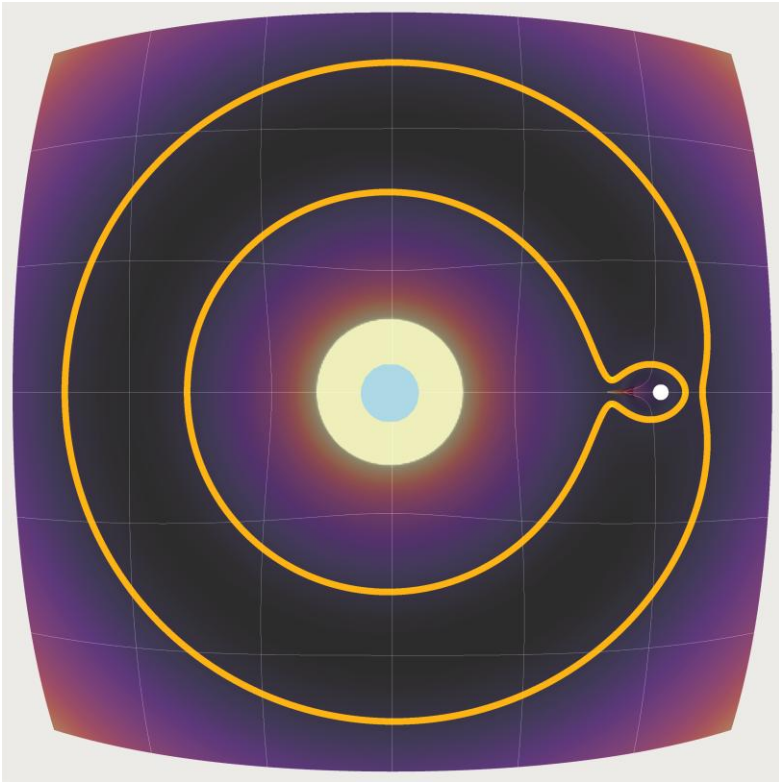
$$\{(x, y, z) : \tilde{U}(x, y, z) \geq C/2\}$$



Hill's Region and Poincaré Section

- Reachable states are bounded by **Hill's Region** :

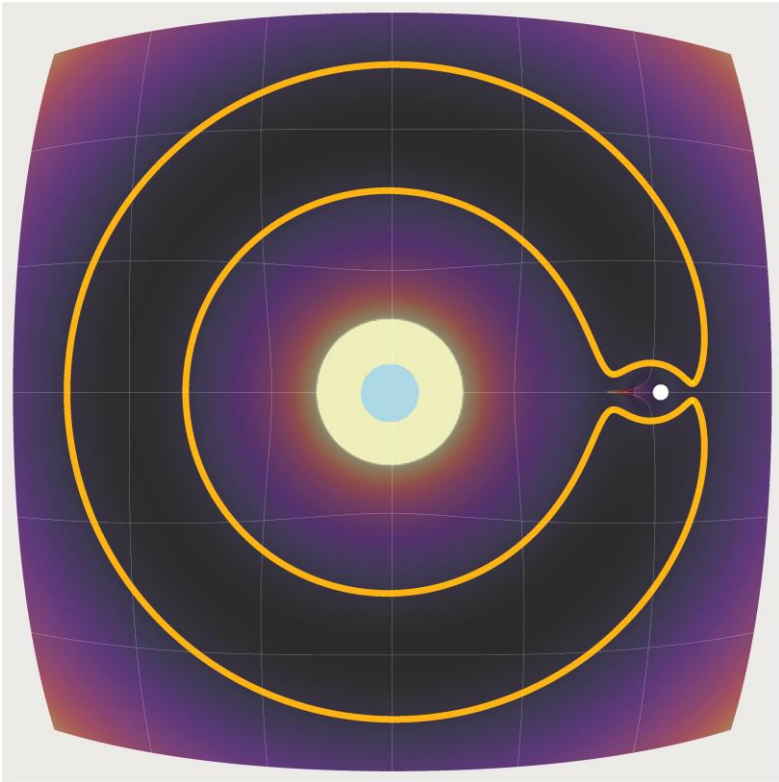
$$\{(x, y, z) : \tilde{U}(x, y, z) \geq C/2\}$$



Hill's Region and Poincaré Section

- Reachable states are bounded by **Hill's Region** :

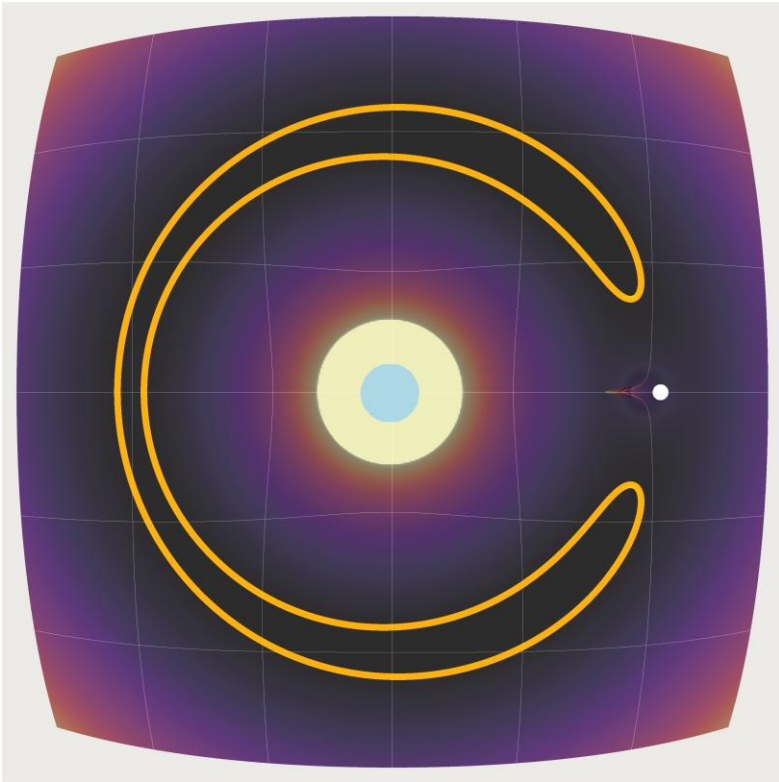
$$\{(x, y, z) : \tilde{U}(x, y, z) \geq C/2\}$$



Hill's Region and Poincaré Section

- Reachable states are bounded by **Hill's Region** :

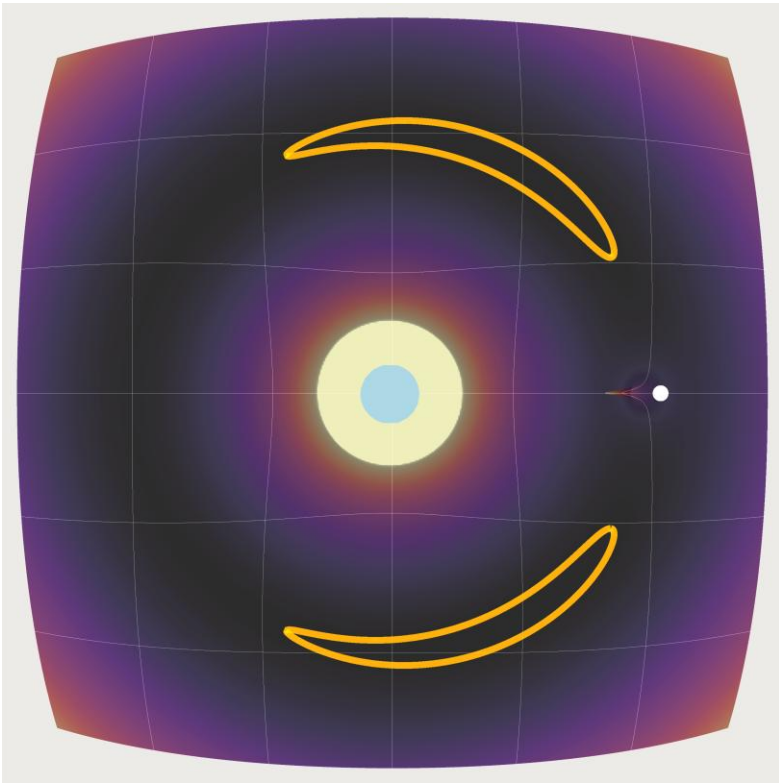
$$\{(x, y, z) : \tilde{U}(x, y, z) \geq C/2\}$$



Hill's Region and Poincaré Section

- Reachable states are bounded by **Hill's Region** :

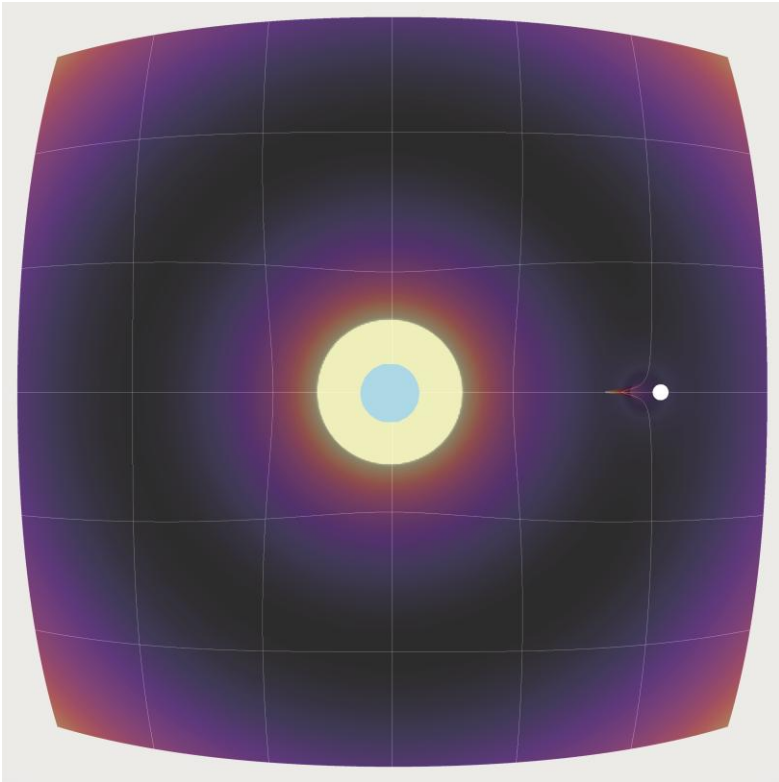
$$\{(x, y, z) : \tilde{U}(x, y, z) \geq C/2\}$$



Hill's Region and Poincaré Section

- Reachable states are bounded by **Hill's Region** :

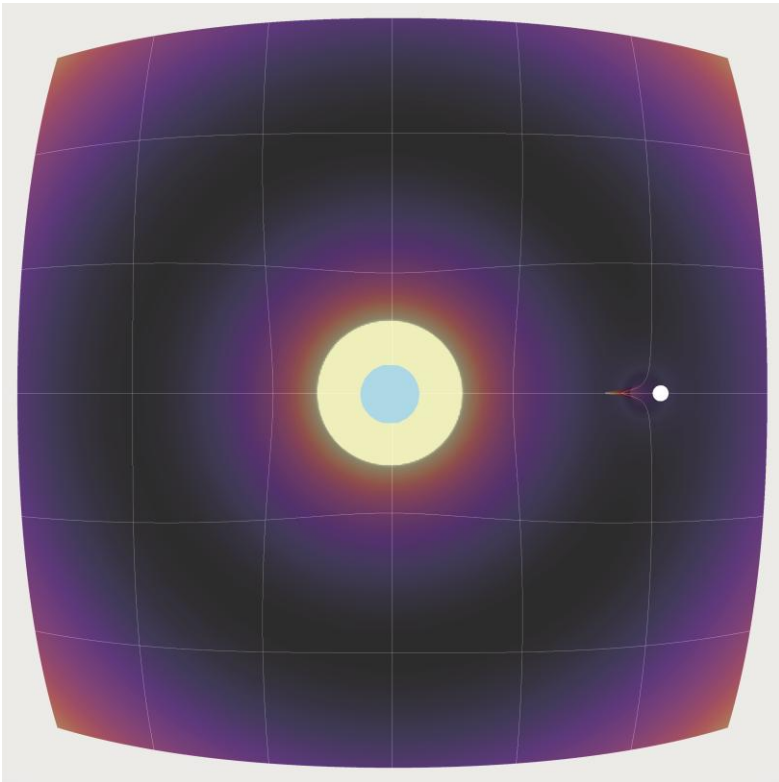
$$\{(x, y, z) : \tilde{U}(x, y, z) \geq C/2\}$$



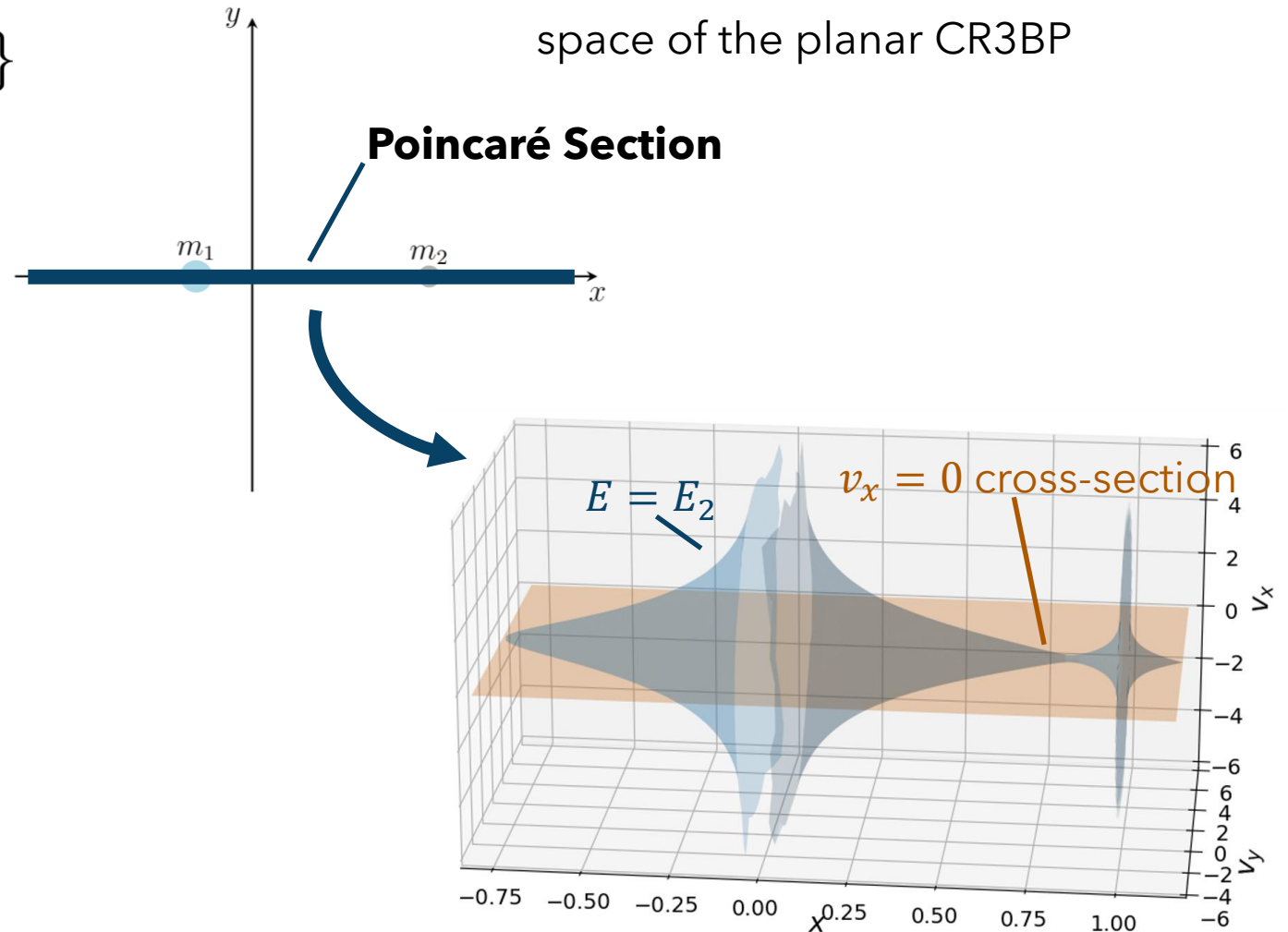
Hill's Region and Poincaré Section

- Reachable states are bounded by **Hill's Region** :

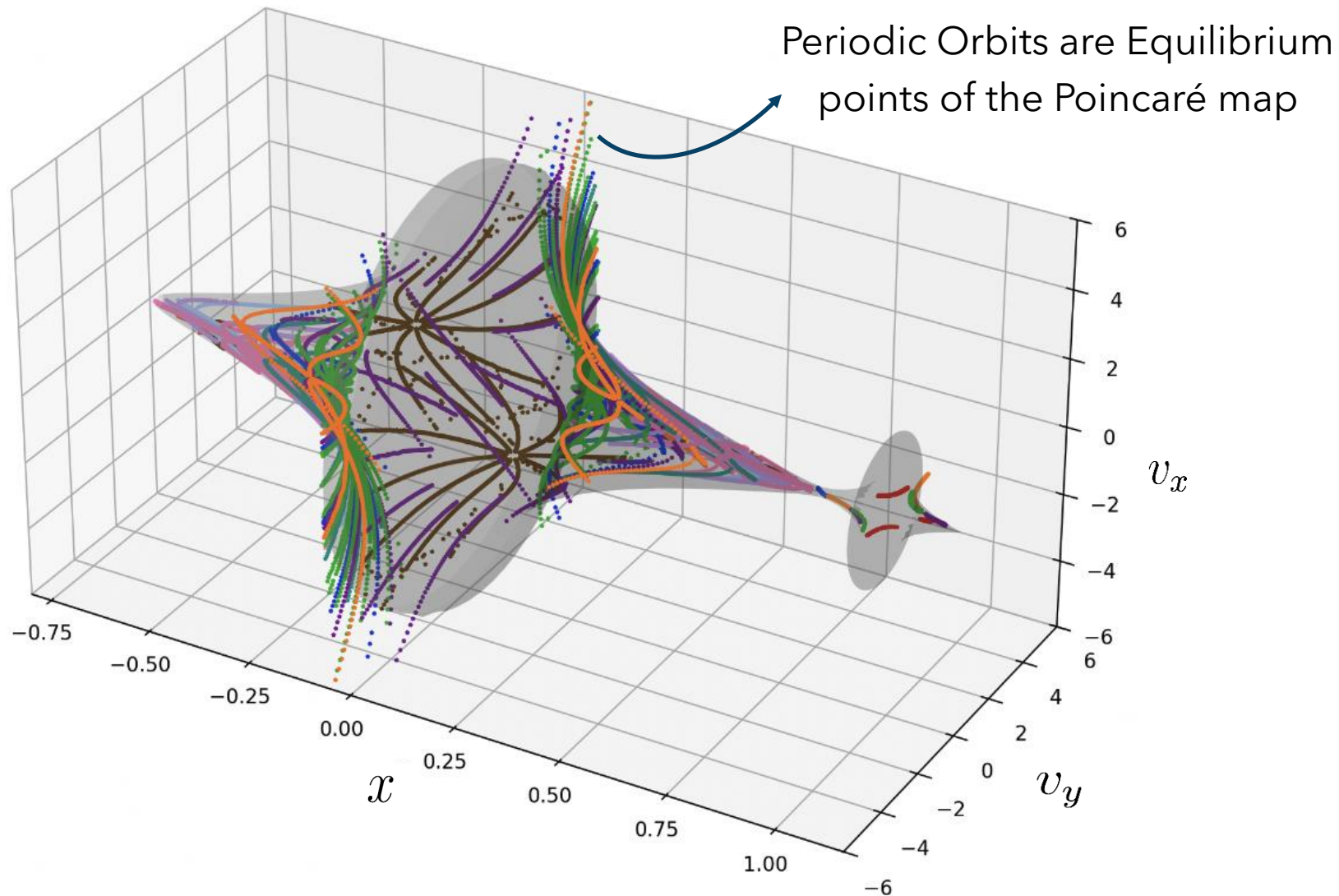
$$\{(x, y, z) : \tilde{U}(x, y, z) \geq C/2\}$$



Visualization in the 4D state space of the planar CR3BP

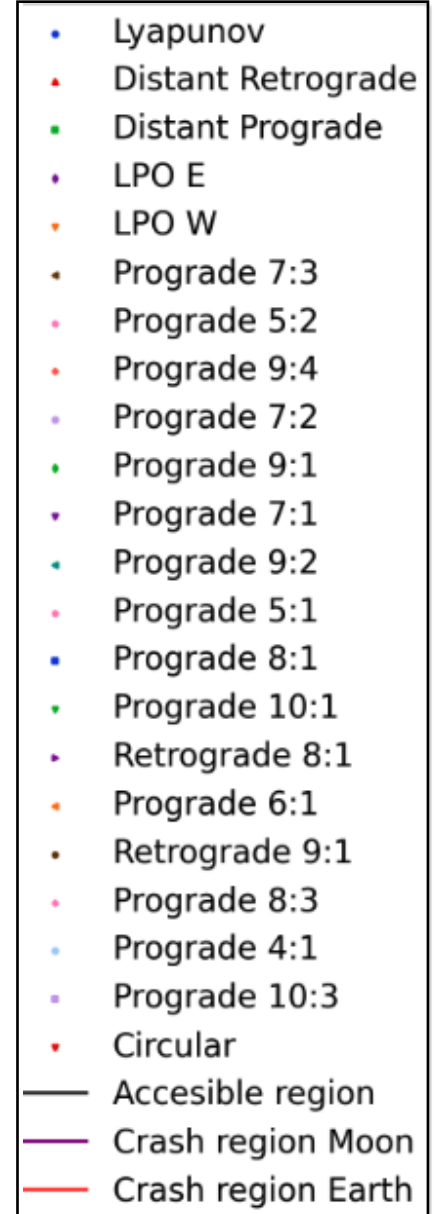
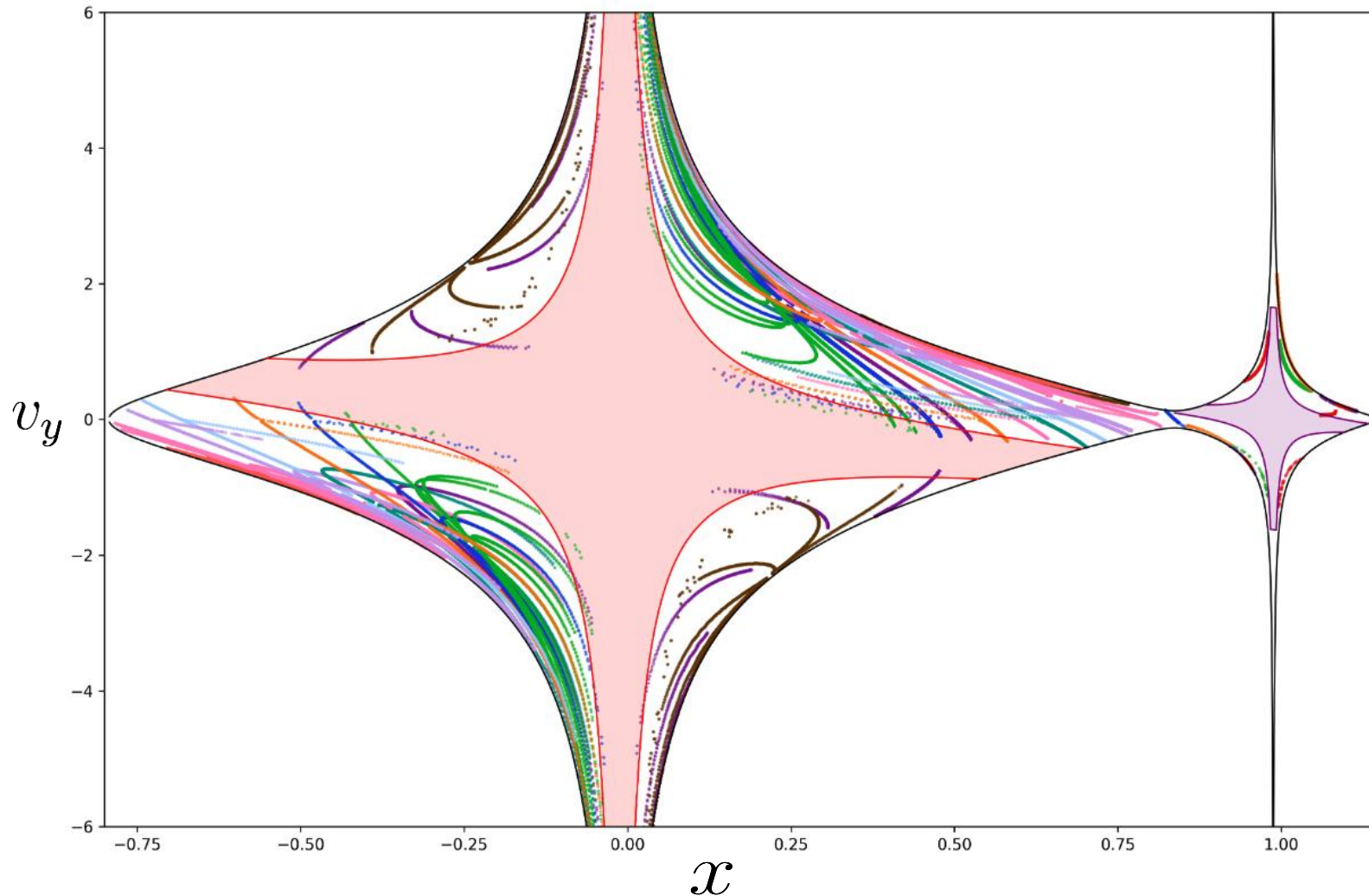


Periodic Orbits on the Poincaré Section

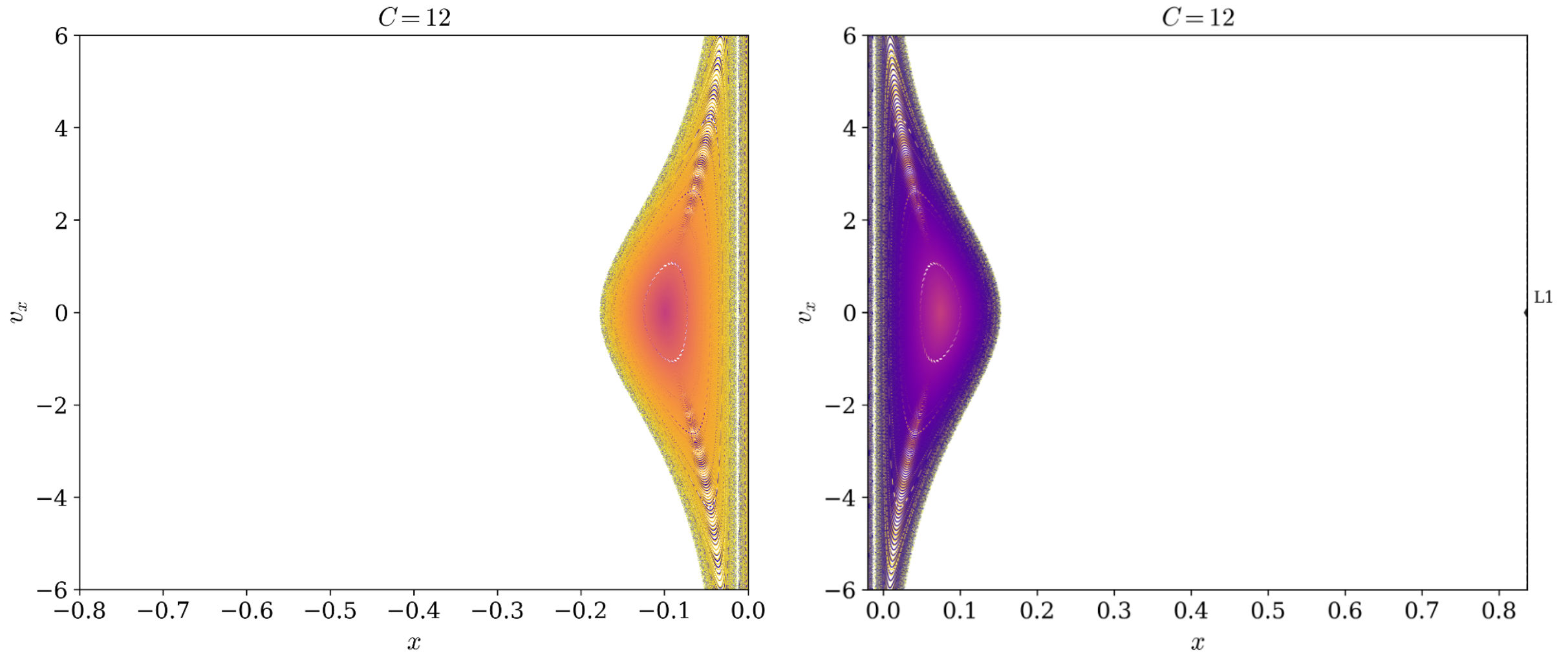


- Lyapunov
- Distant Retrograde
- Distant Prograde
- LPO E
- LPO W
- Prograde 7:3
- Prograde 5:2
- Prograde 9:4
- Prograde 7:2
- Prograde 9:1
- Prograde 7:1
- Prograde 9:2
- Prograde 5:1
- Prograde 8:1
- Prograde 10:1
- Retrograde 8:1
- Prograde 6:1
- Retrograde 9:1
- Prograde 8:3
- Prograde 4:1
- Prograde 10:3
- Circular
- Accessible region
- Crash region Moon
- Crash region Earth

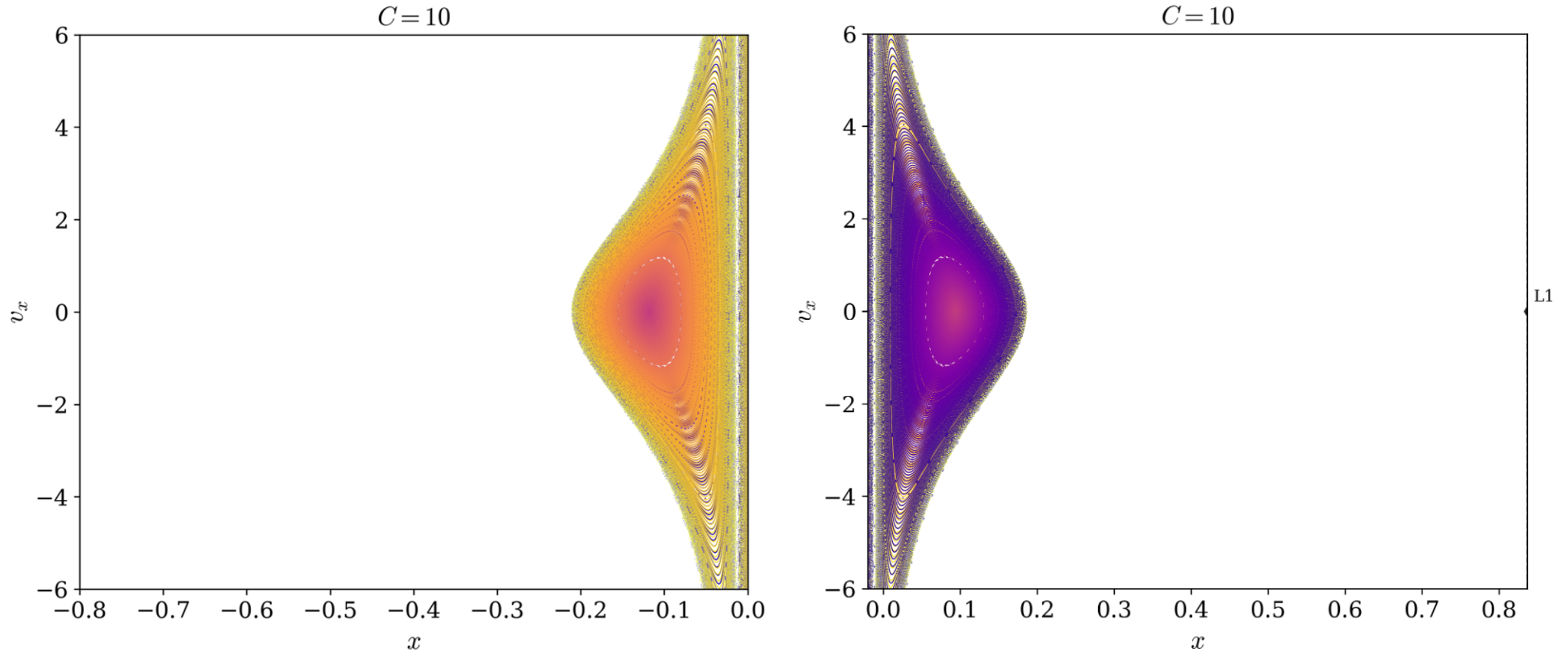
Periodic Orbits on the Poincaré Section



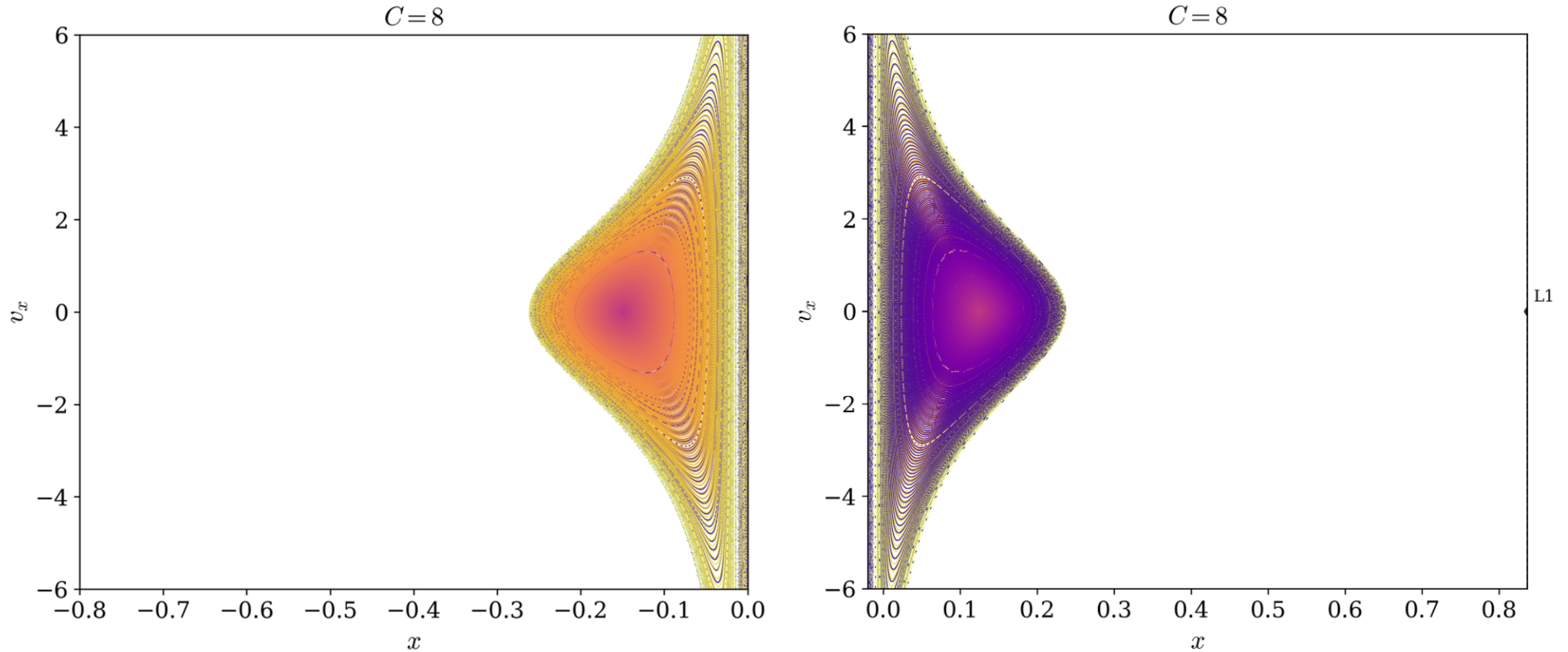
Constant energy Poincaré maps



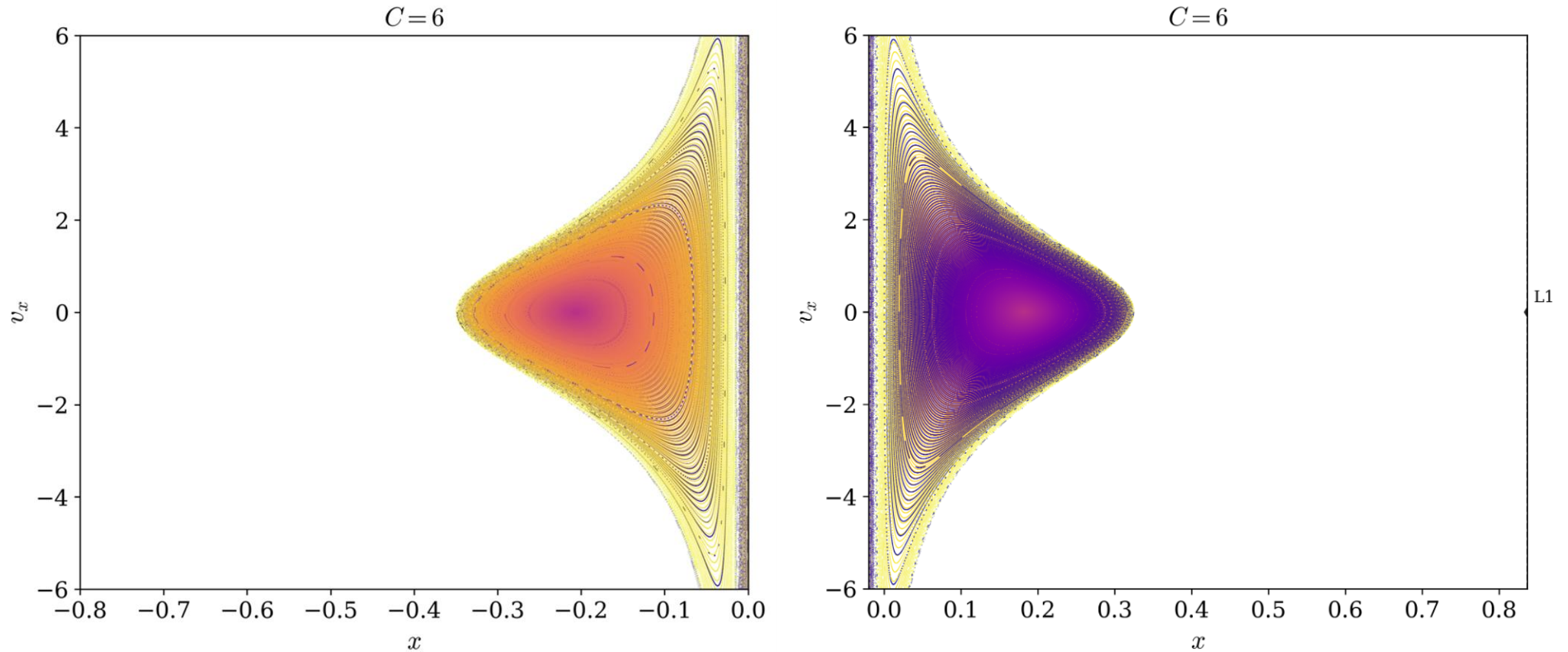
Constant energy Poincaré maps



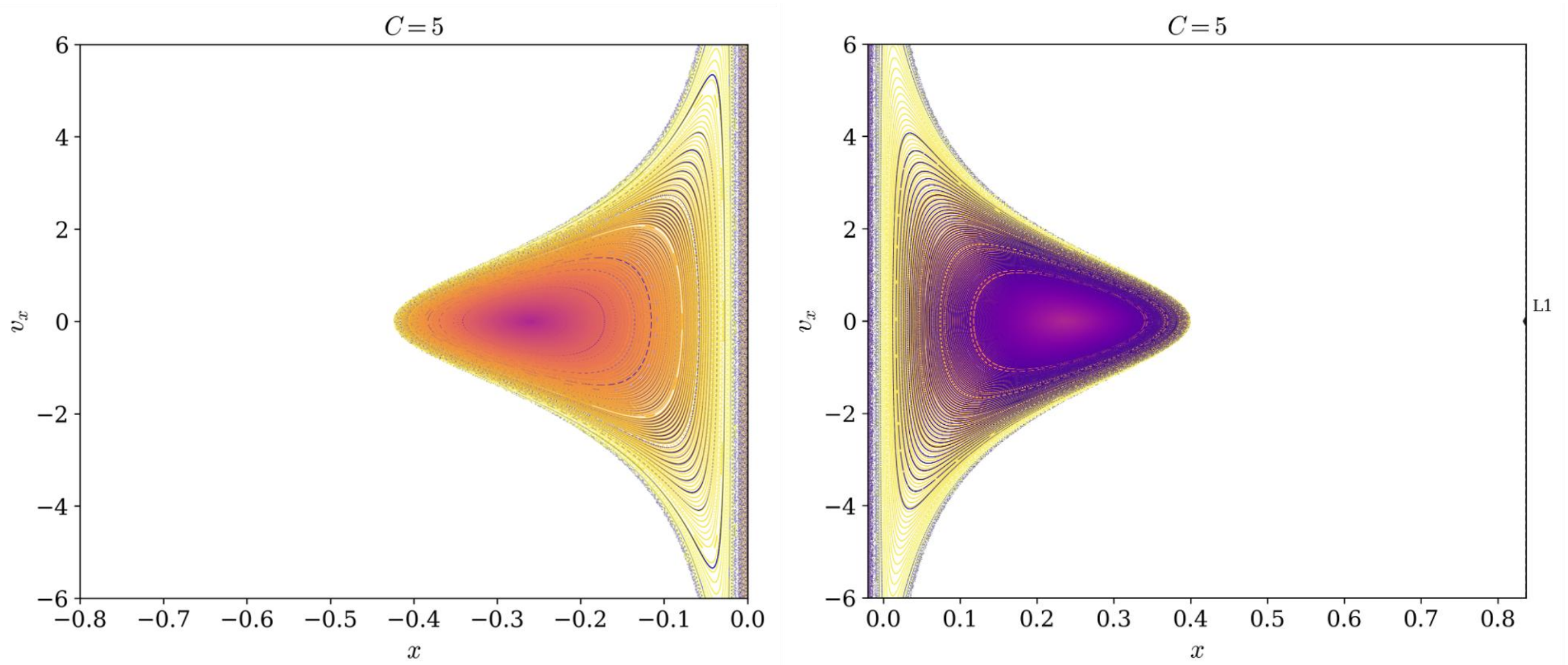
Constant energy Poincaré maps



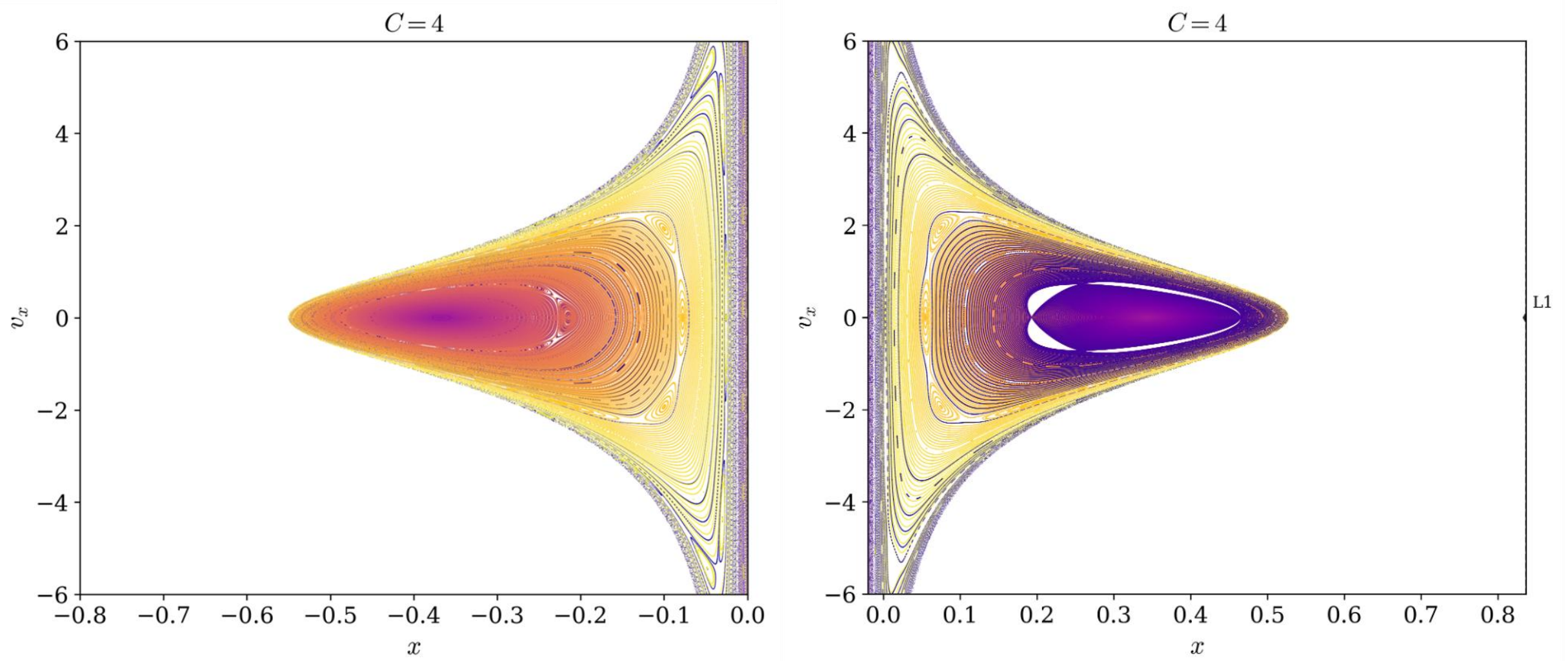
Constant energy Poincaré maps



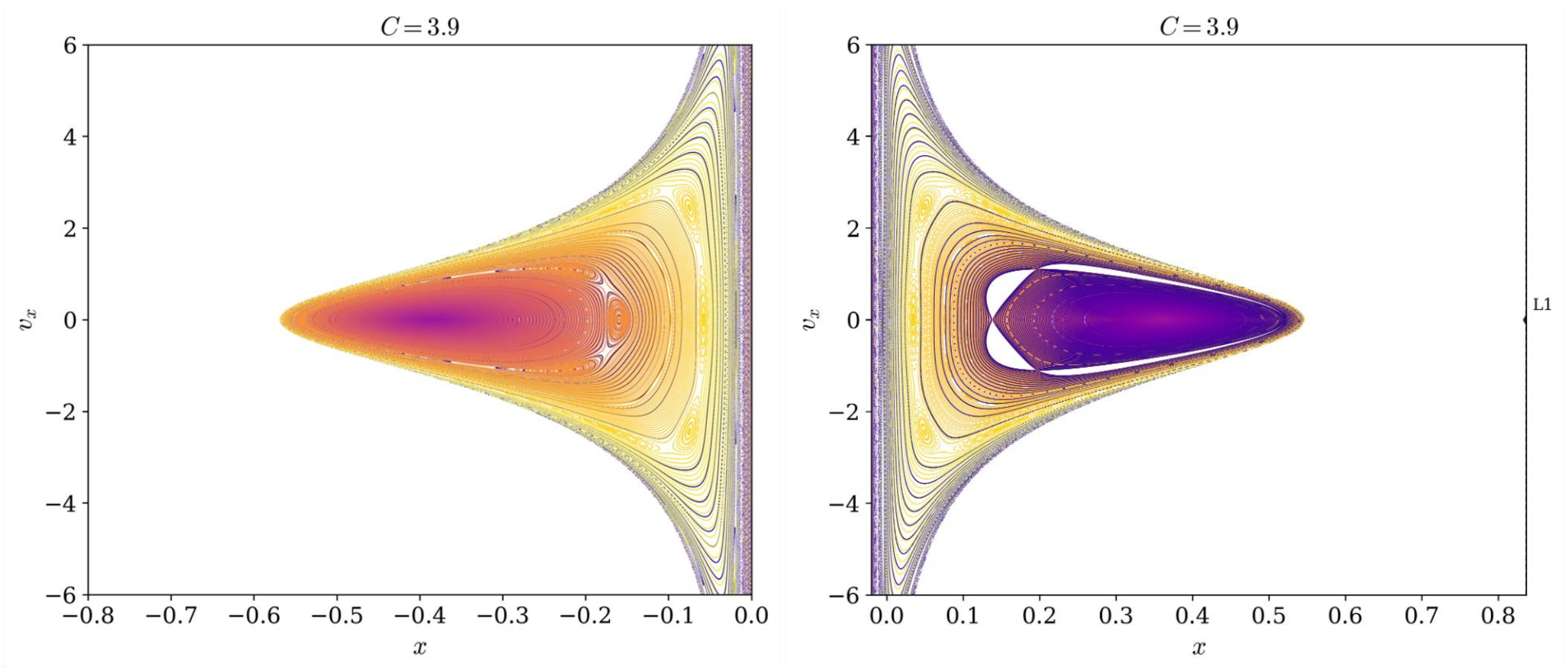
Constant energy Poincaré maps



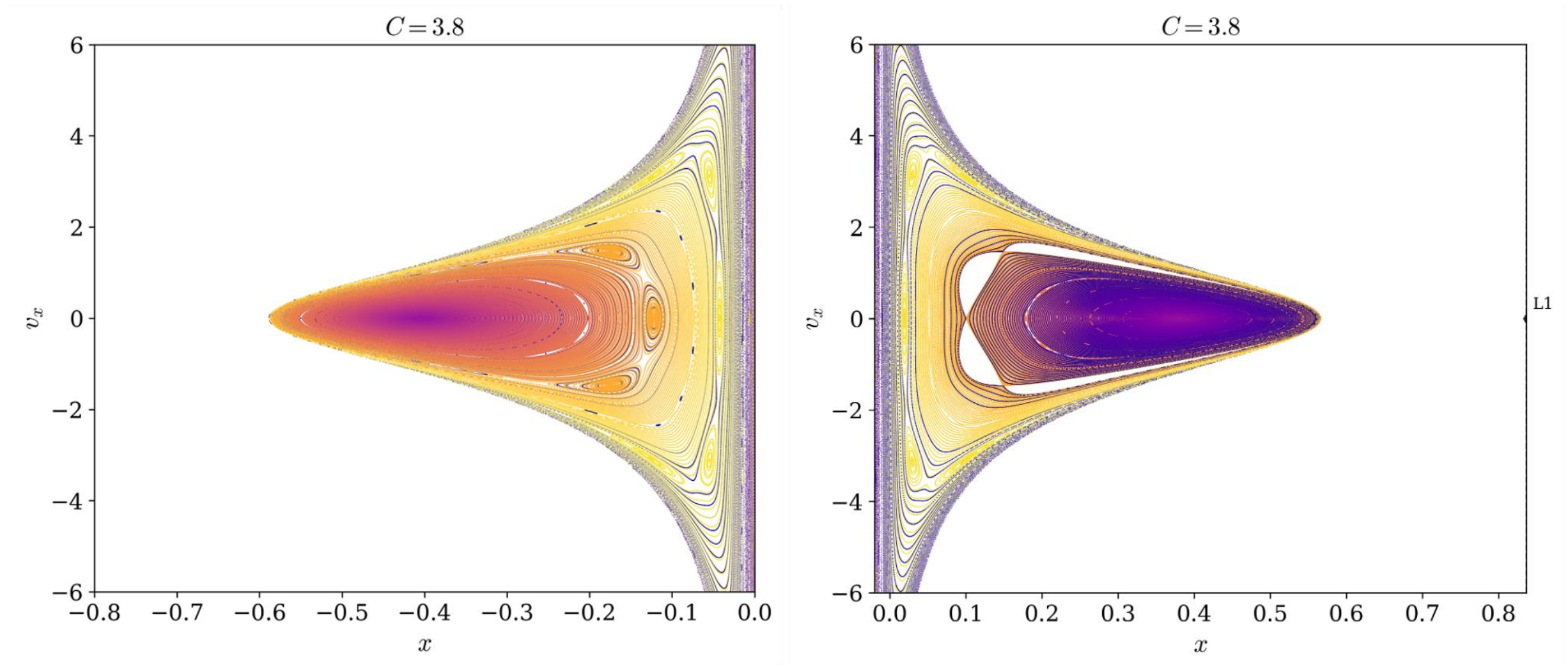
Constant energy Poincaré maps



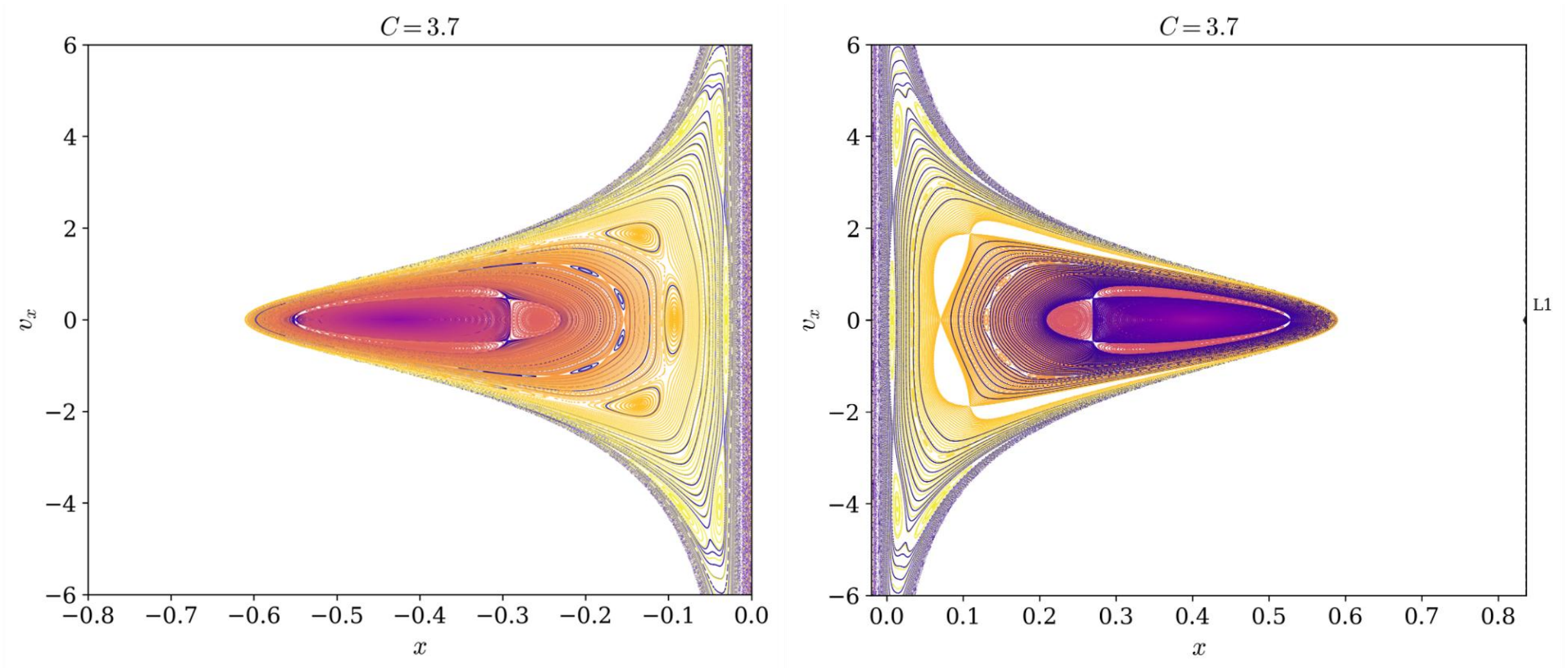
Constant energy Poincaré maps



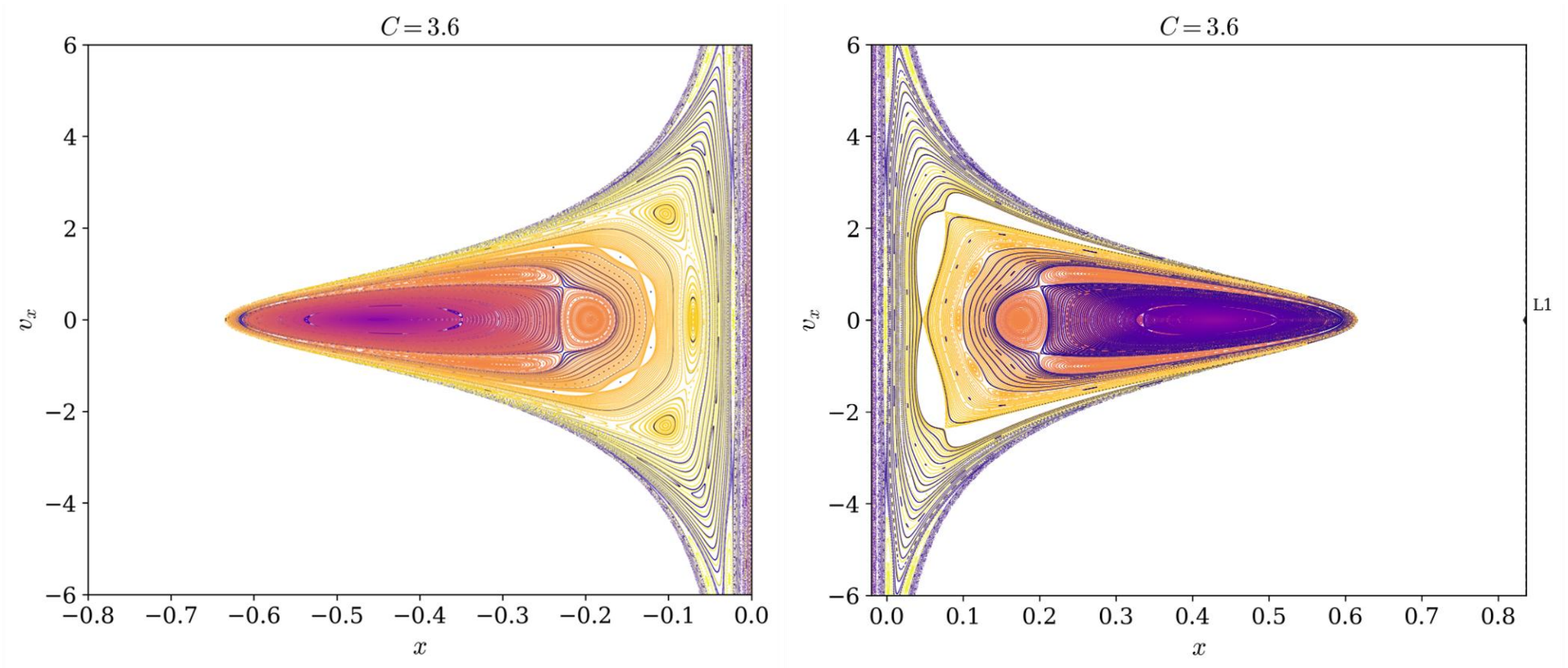
Constant energy Poincaré maps



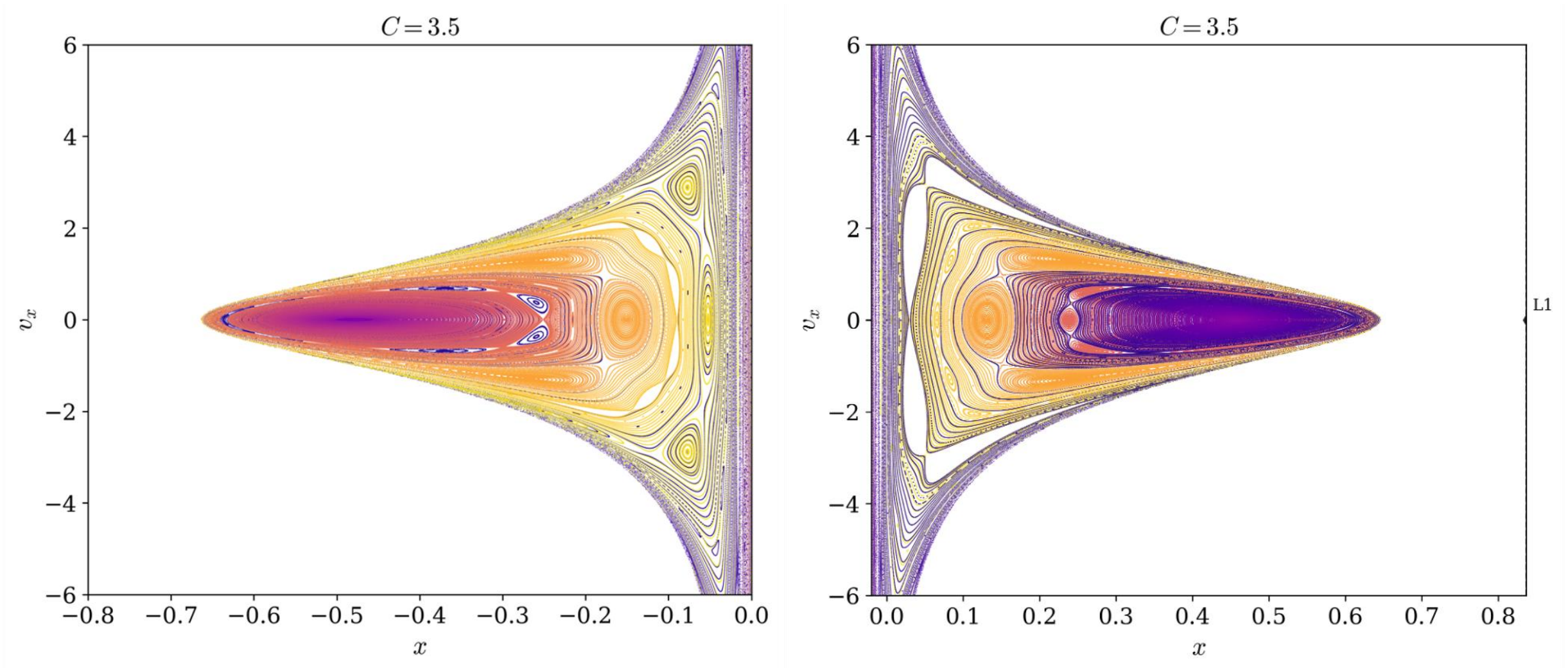
Constant energy Poincaré maps



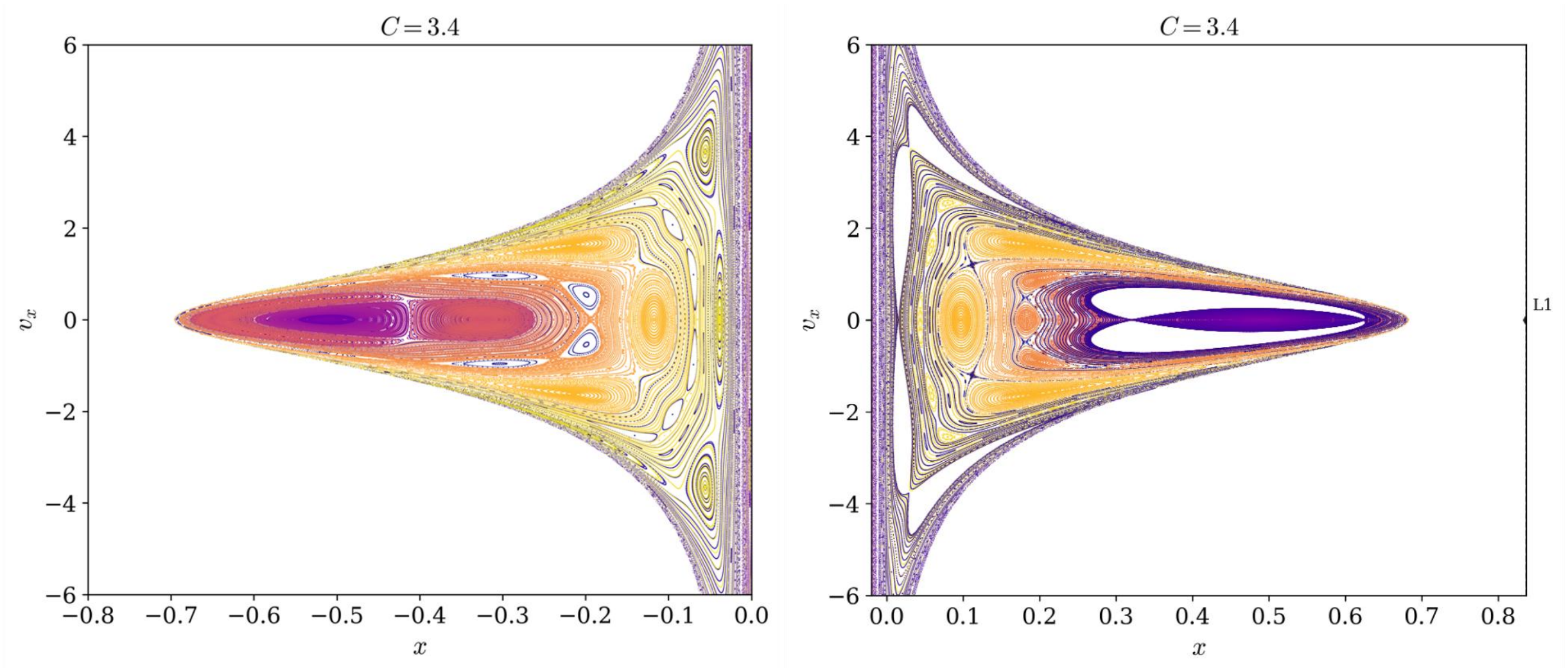
Constant energy Poincaré maps



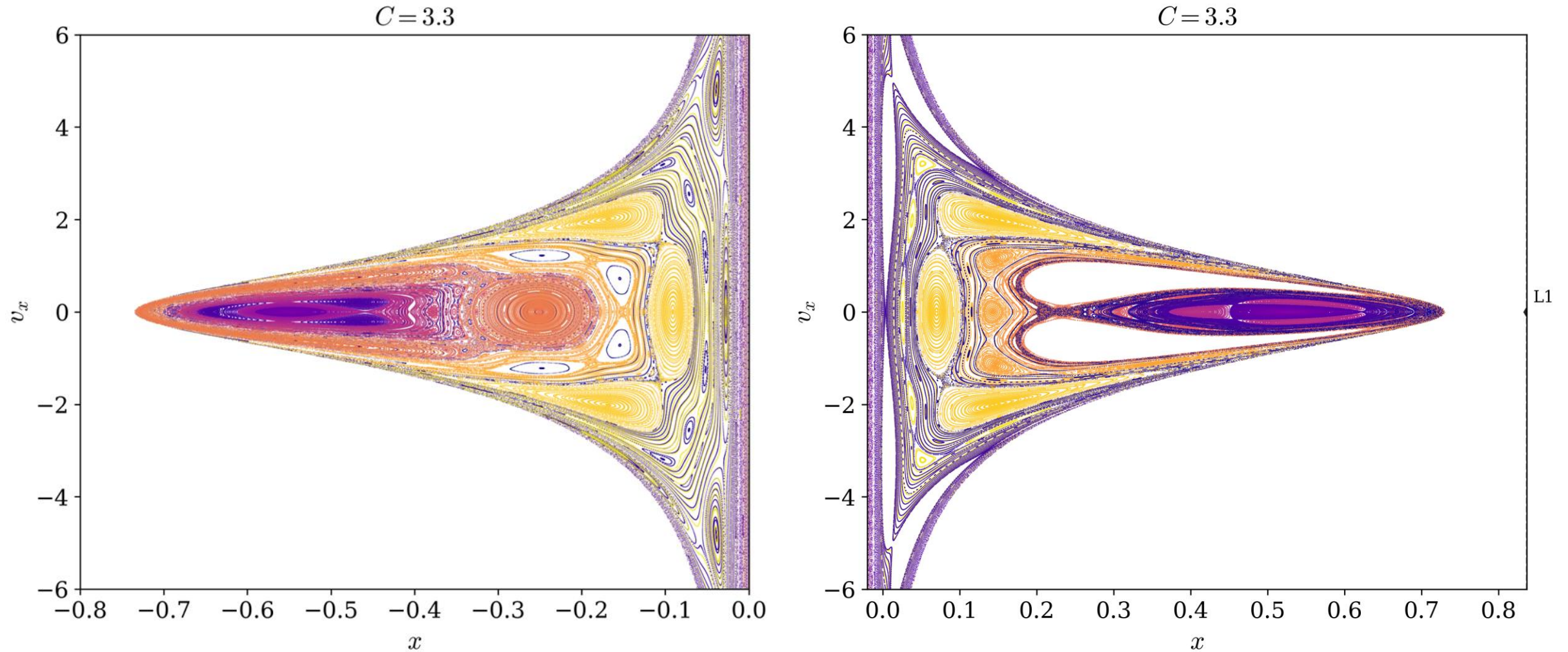
Constant energy Poincaré maps



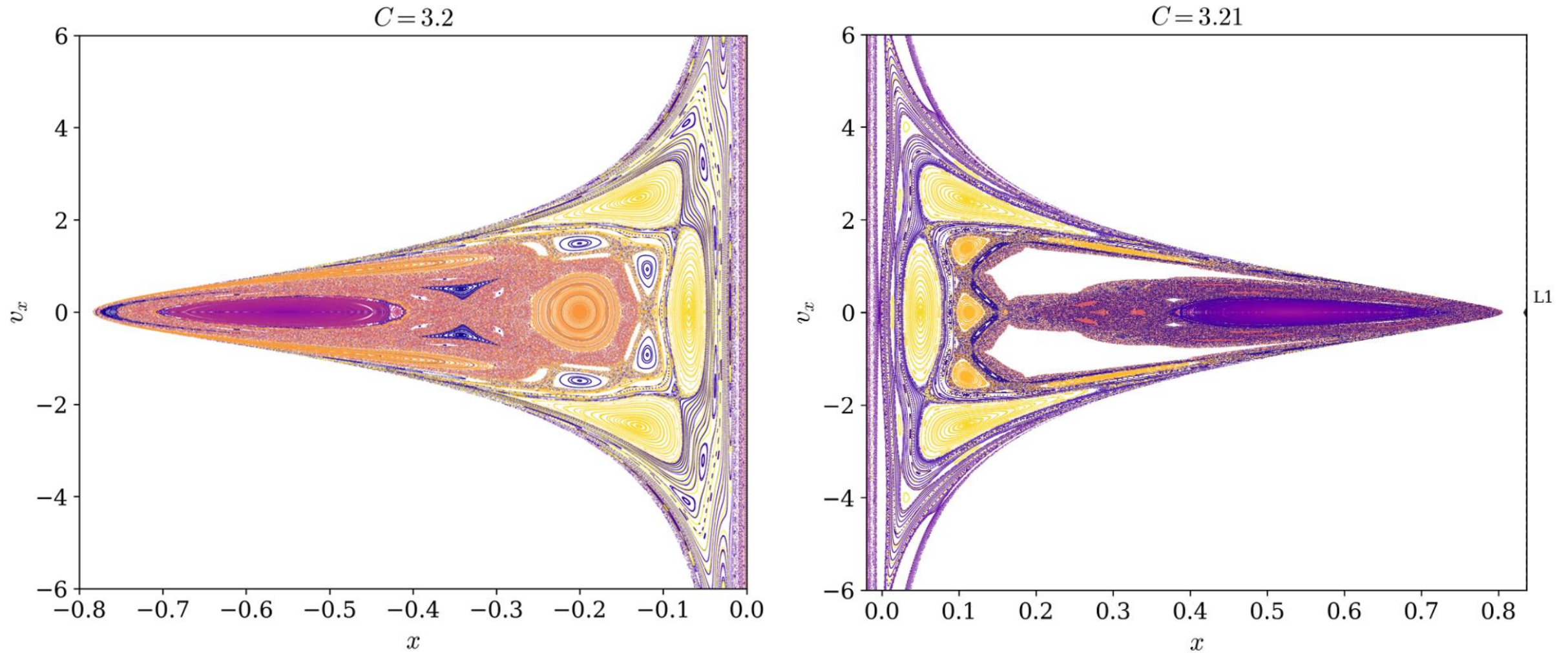
Constant energy Poincaré maps



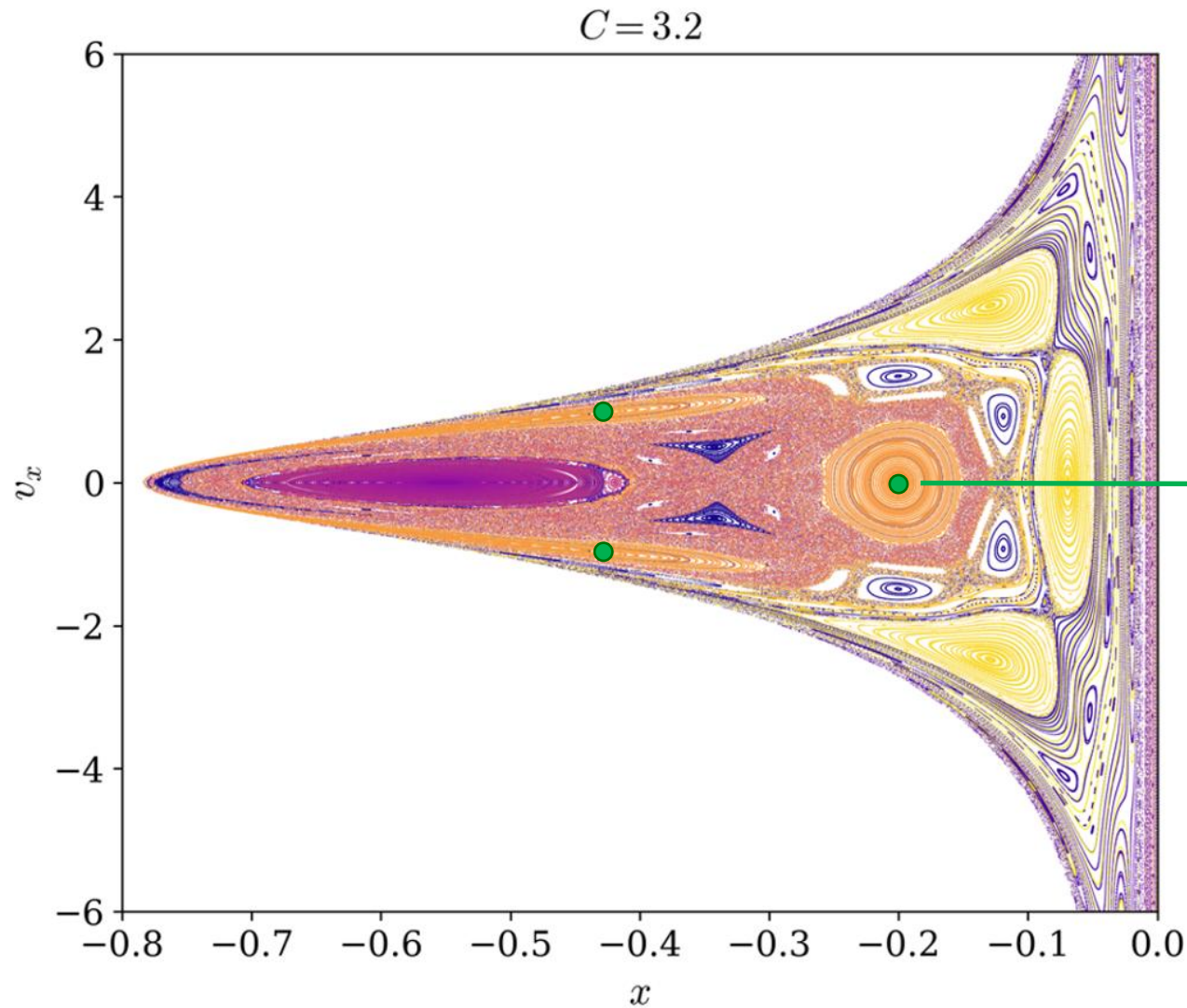
Constant energy Poincaré maps



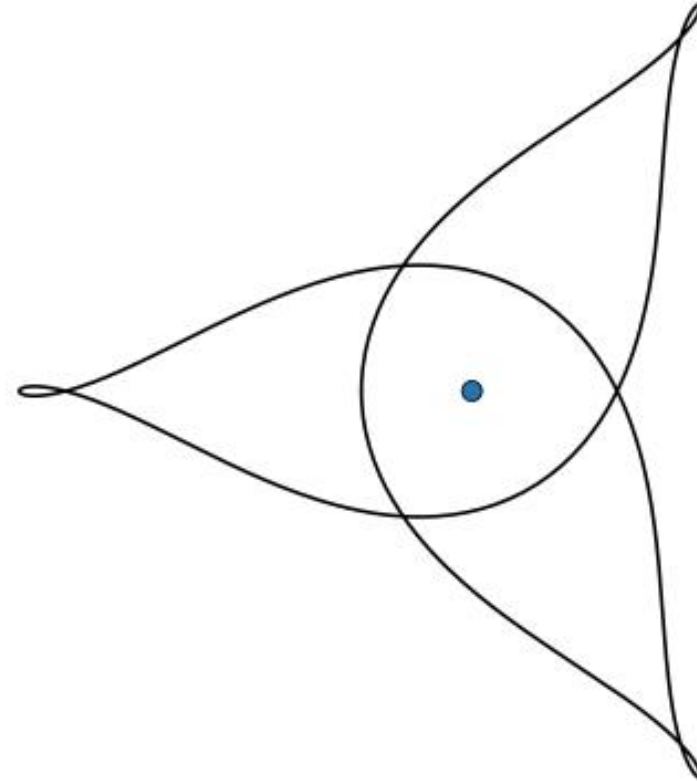
Constant energy Poincaré maps



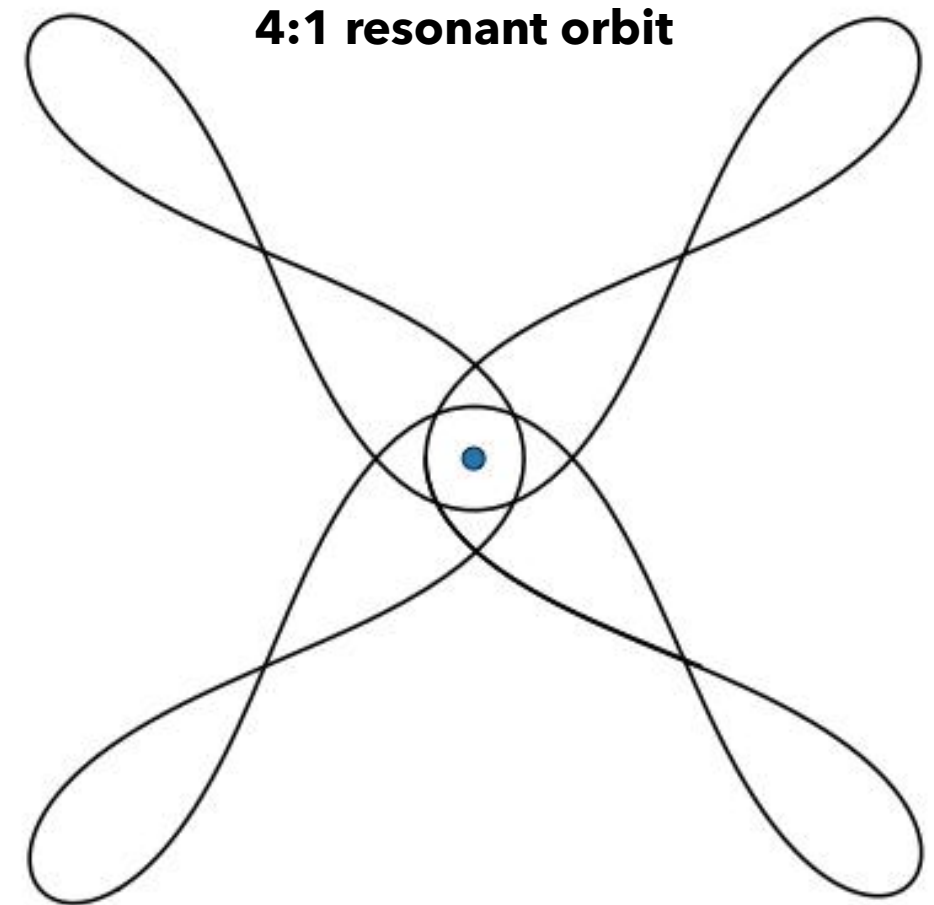
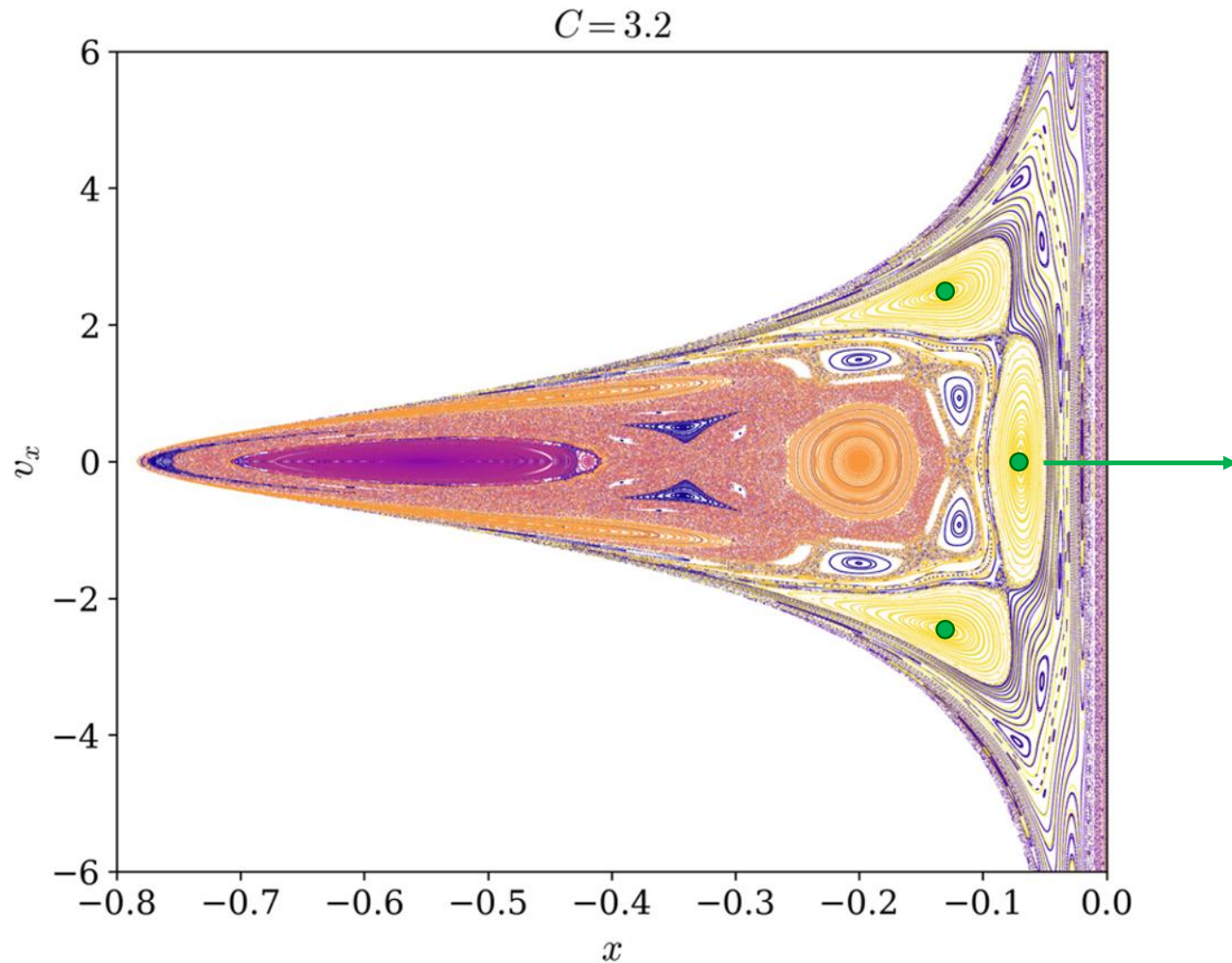
Islands correspond to Resonant Orbits



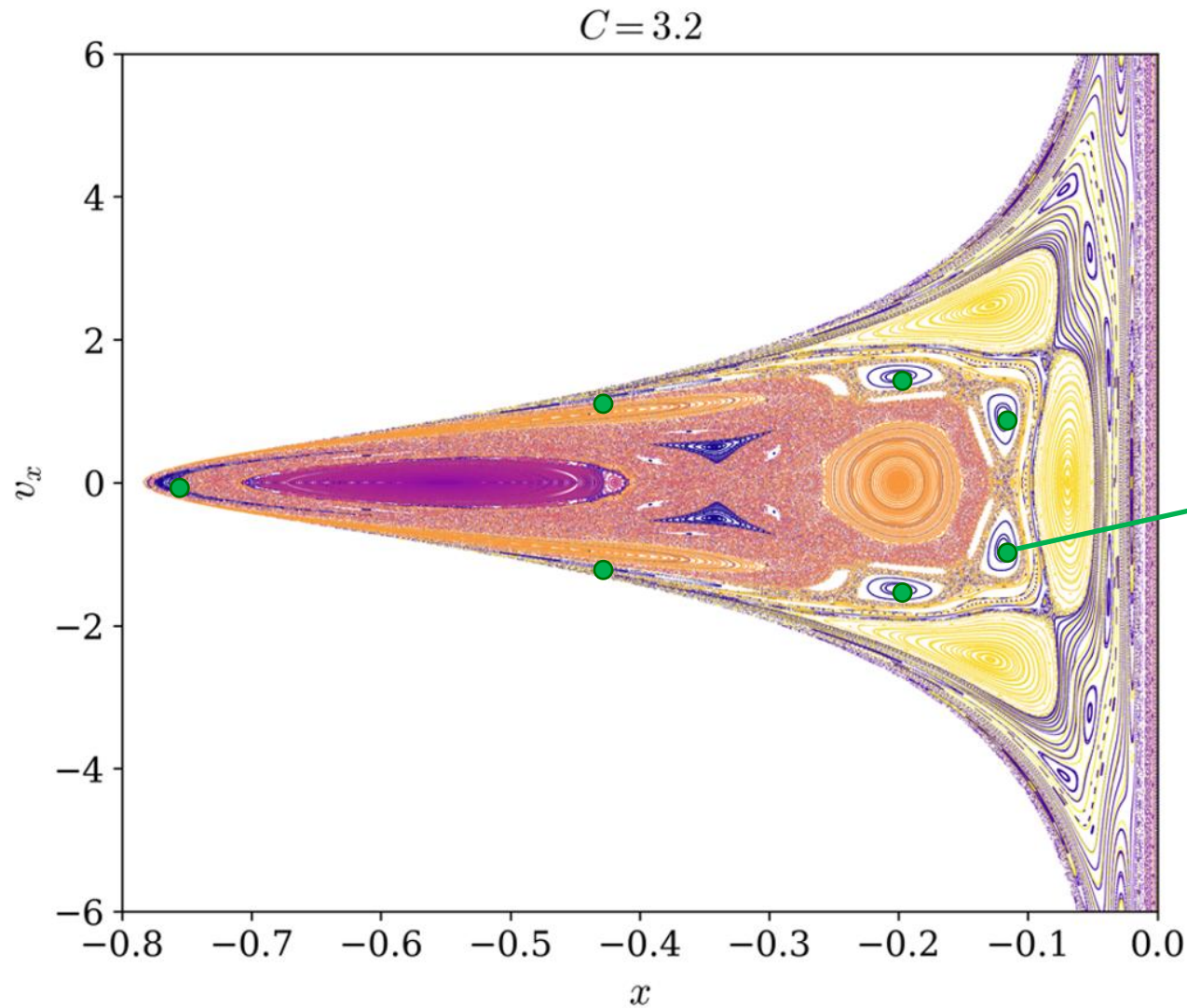
3:1 resonant orbit



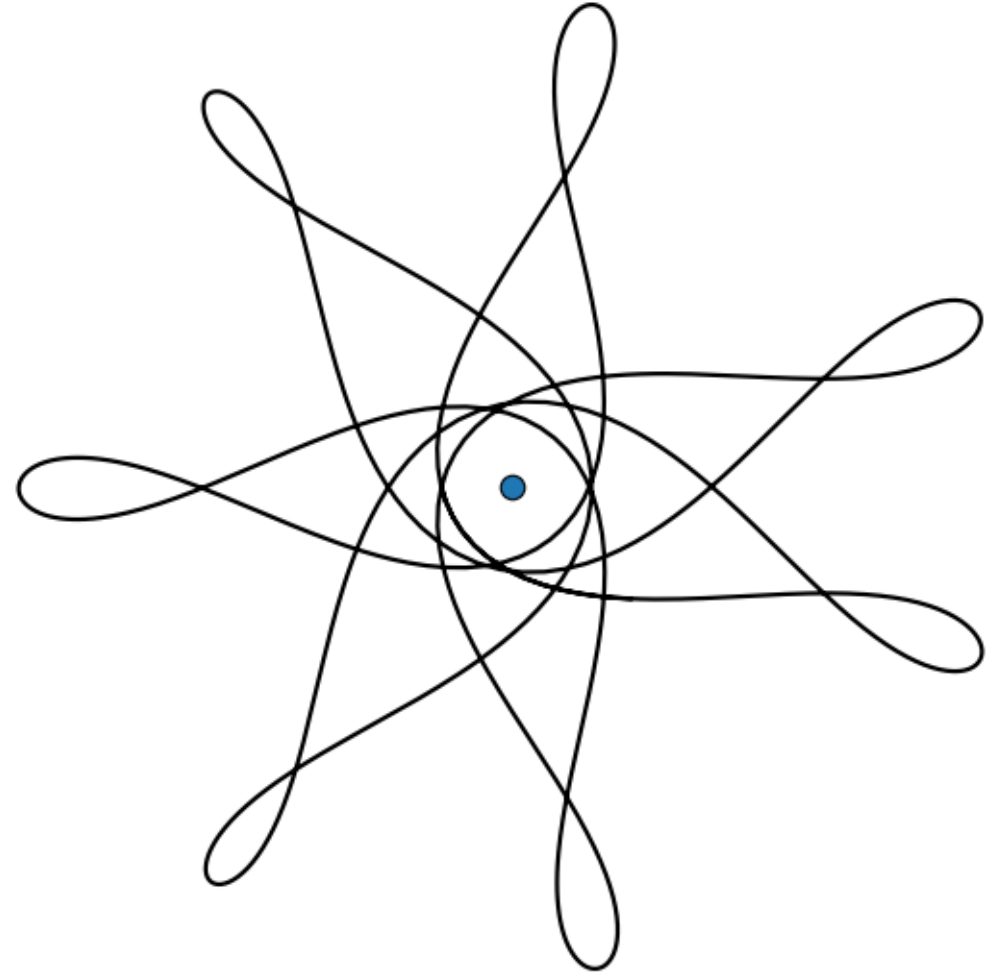
Islands correspond to Resonant Orbits



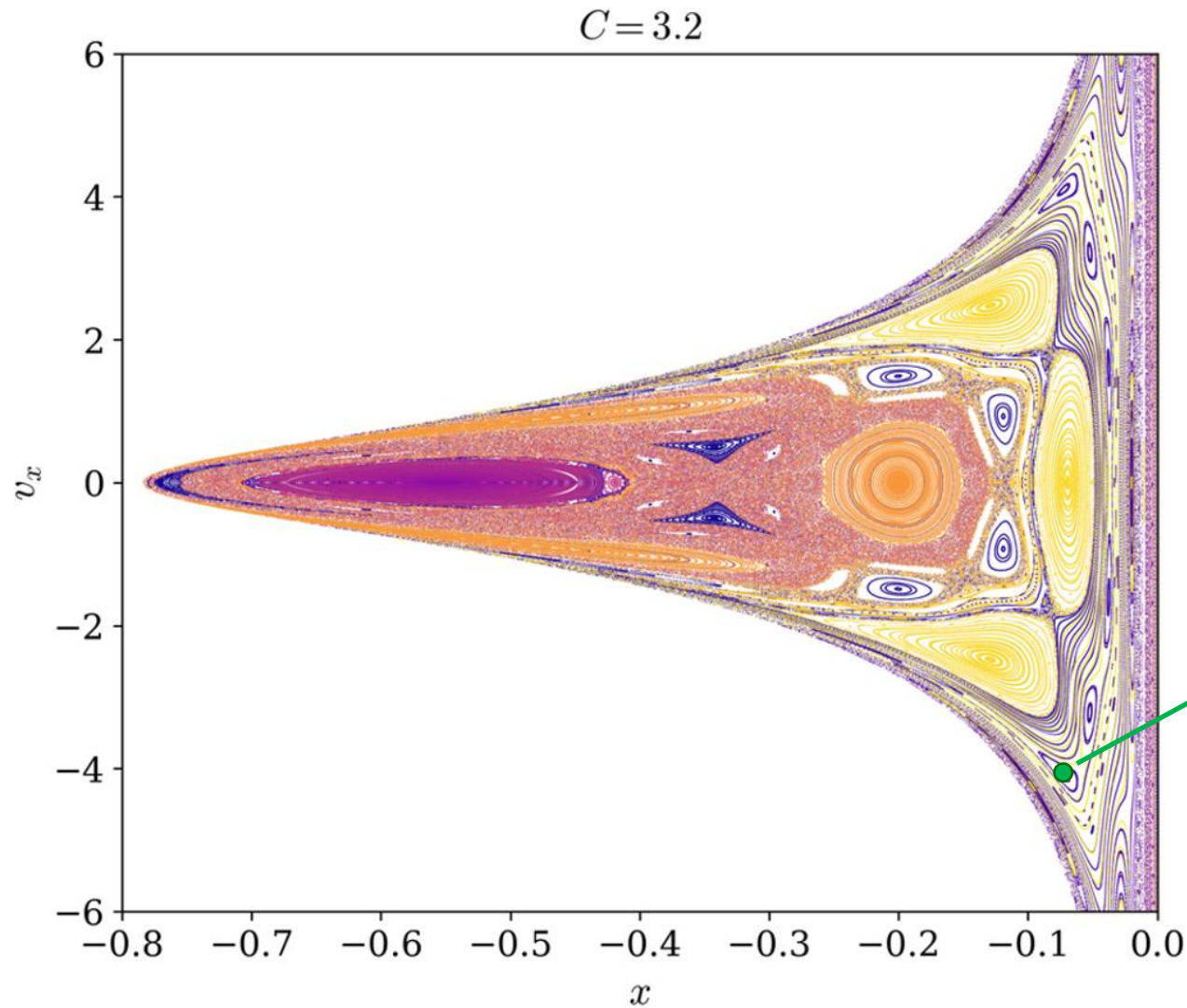
Islands correspond to Resonant Orbits



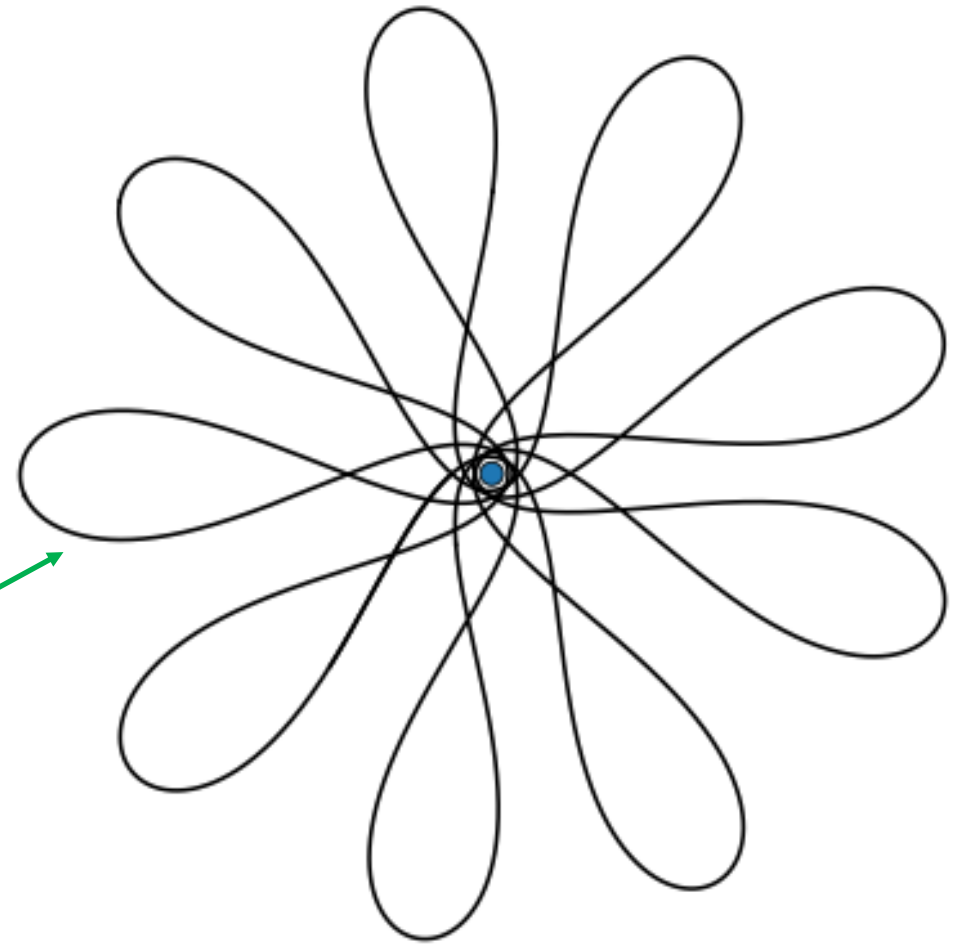
7:2 resonant orbit



Islands correspond to Resonant Orbits



9:2 resonant orbit



Normal Forms around Periodic Orbits

- Simplified dynamics near a periodic orbit through a series of **near-identical transformations**

Original Hamiltonian Flow

$$\begin{aligned}\dot{\xi} &= J\nabla H(\xi) \\ \dot{\xi}_\gamma &= J\nabla H(\xi_\gamma) \\ \gamma(t+T) &= \gamma(t) \\ J &\text{ symplectic}\end{aligned}$$

Taylor-Fourier Expansion

$$H_d^{(\eta)}(t, \eta) = \sum_{k \in \mathbb{Z}} H_{d,k}(\eta) e^{ik\omega t}, \quad \omega = 2\pi/T$$

Degree-by-degree normalization

At degree d

$$K^{\text{cur}}(t, \eta) = H_2(\eta) + \sum_{m=3}^{d-1} Z_m(t, \eta) + H_d^{\text{cur}}(t, \eta) + O(\|\eta\|^{d+1})$$

Pick generating function

$$\mathcal{L}_d W_d + Z_d = H_d^{\text{cur}}, \quad \mathcal{L}_d W_d := \partial_t W_d + \{W, H_2\}$$

Transformed Hamiltonian

$$K^{\text{new}}(t, \eta) = H_2(\eta) + \sum_{m=3}^d Z_m(t, \eta) + O(\|\eta\|^{d+1})$$

Symplectic Floquet Reduction

$$\xi' = P(t)\eta \quad \Phi(t) = P(t)e^{tB}$$

$$K(t, \eta) = H_2(\eta) + \sum_{d \geq 3} H_d^{(\eta)}(t, \eta),$$

$$H_2(\eta) = \frac{1}{2} \eta^\top (-JB) \eta$$

$$H_d^{(\eta)}(t, \eta) \text{ homogenous polynomial of degree } d \text{ in } \eta$$

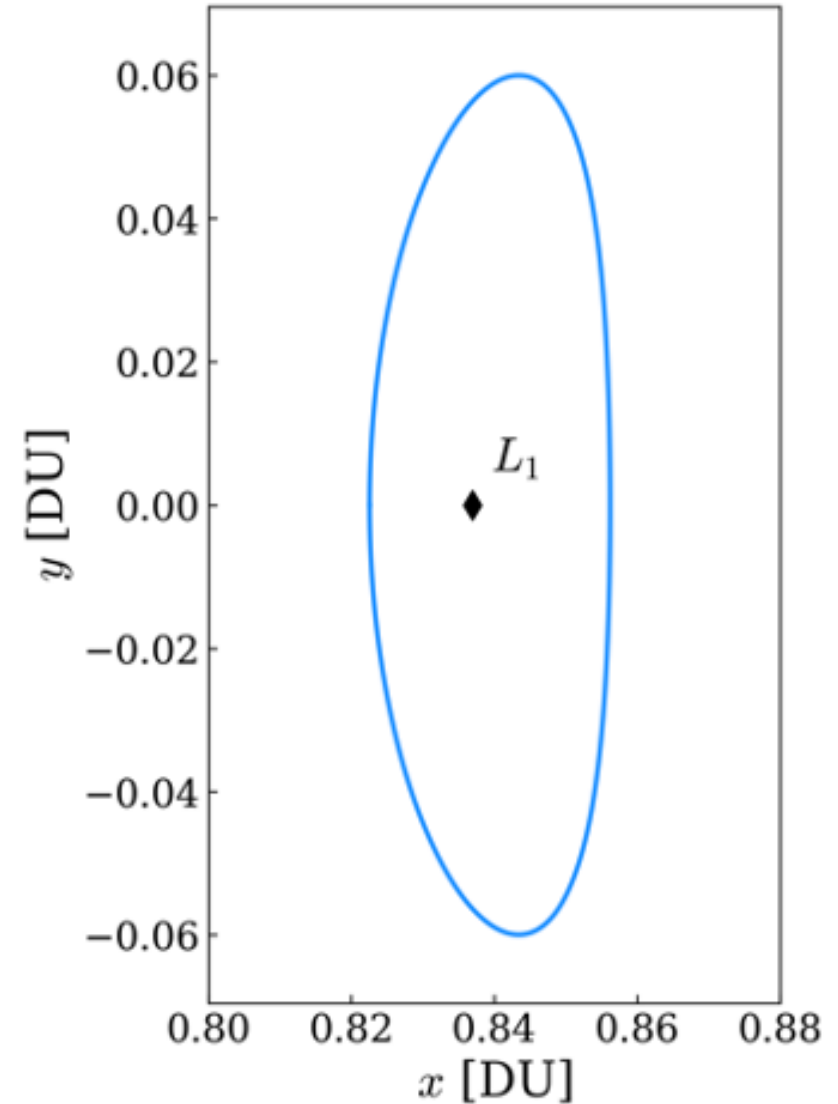
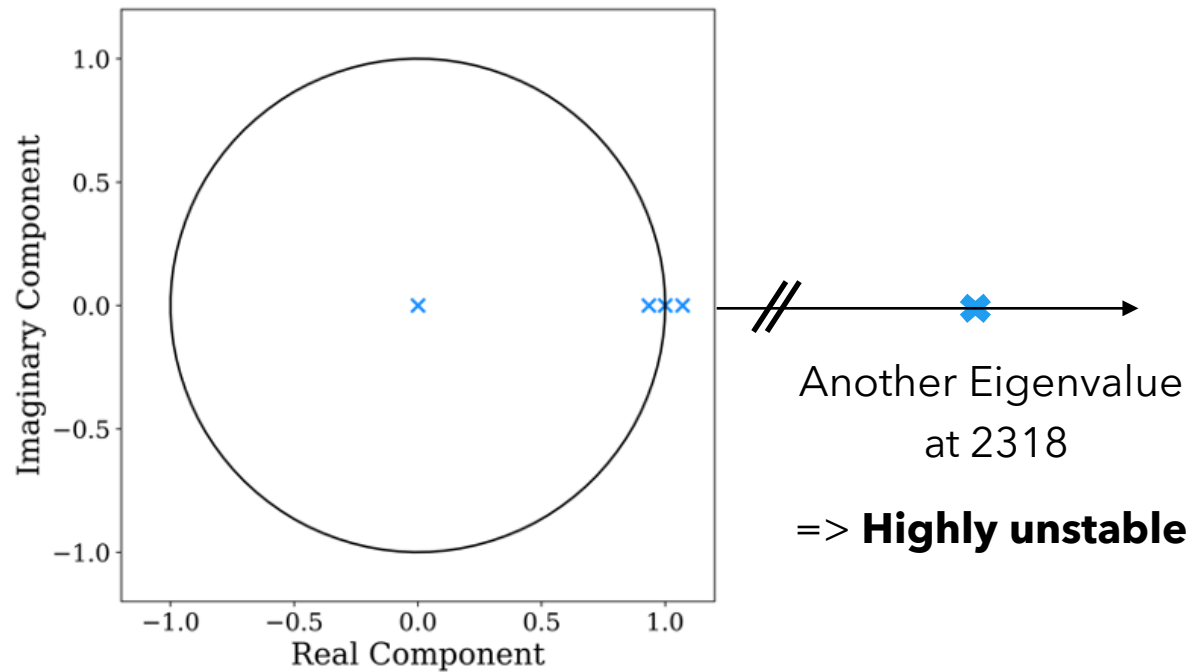
Coordinate Translation

$$\xi' := \xi - \xi_\gamma$$

Normal Forms around Periodic Orbits

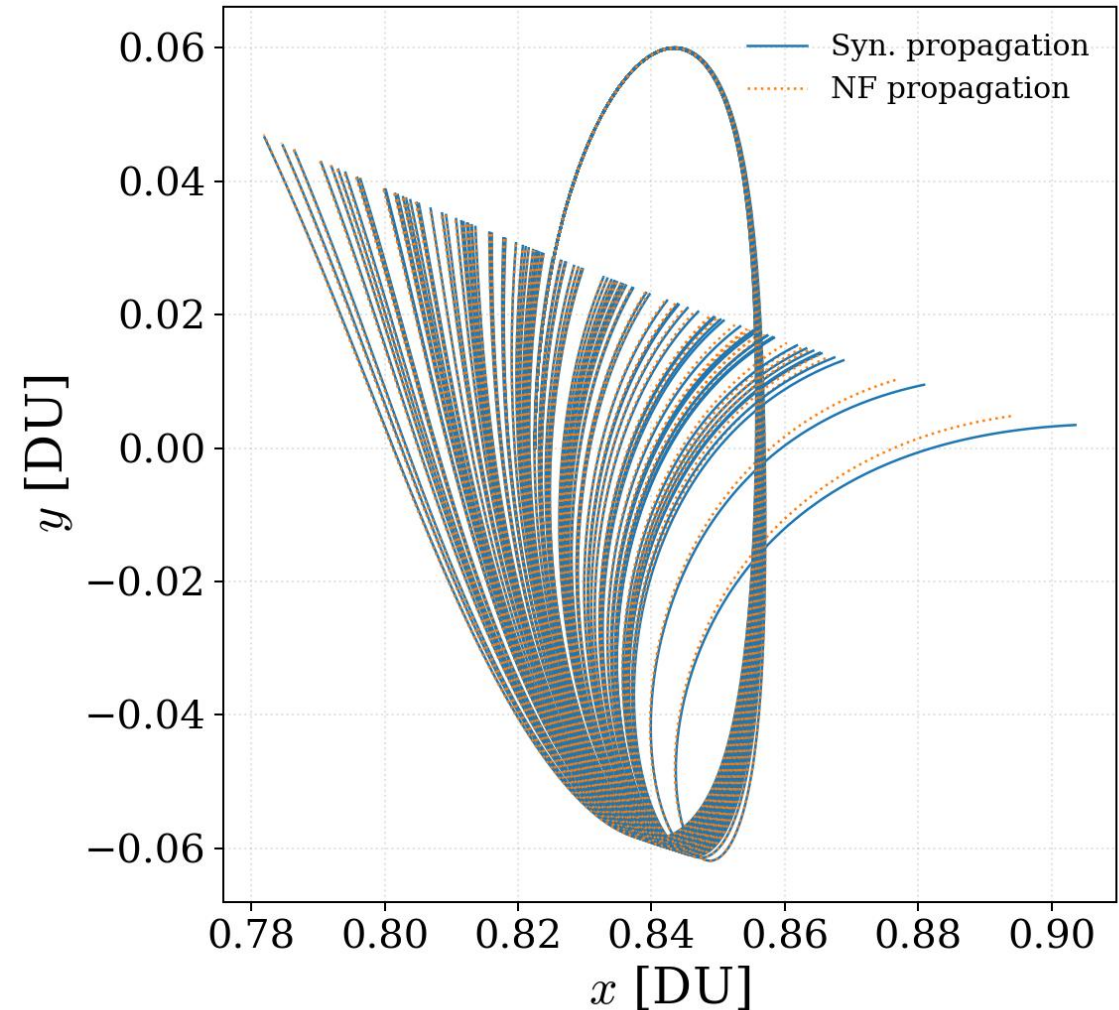
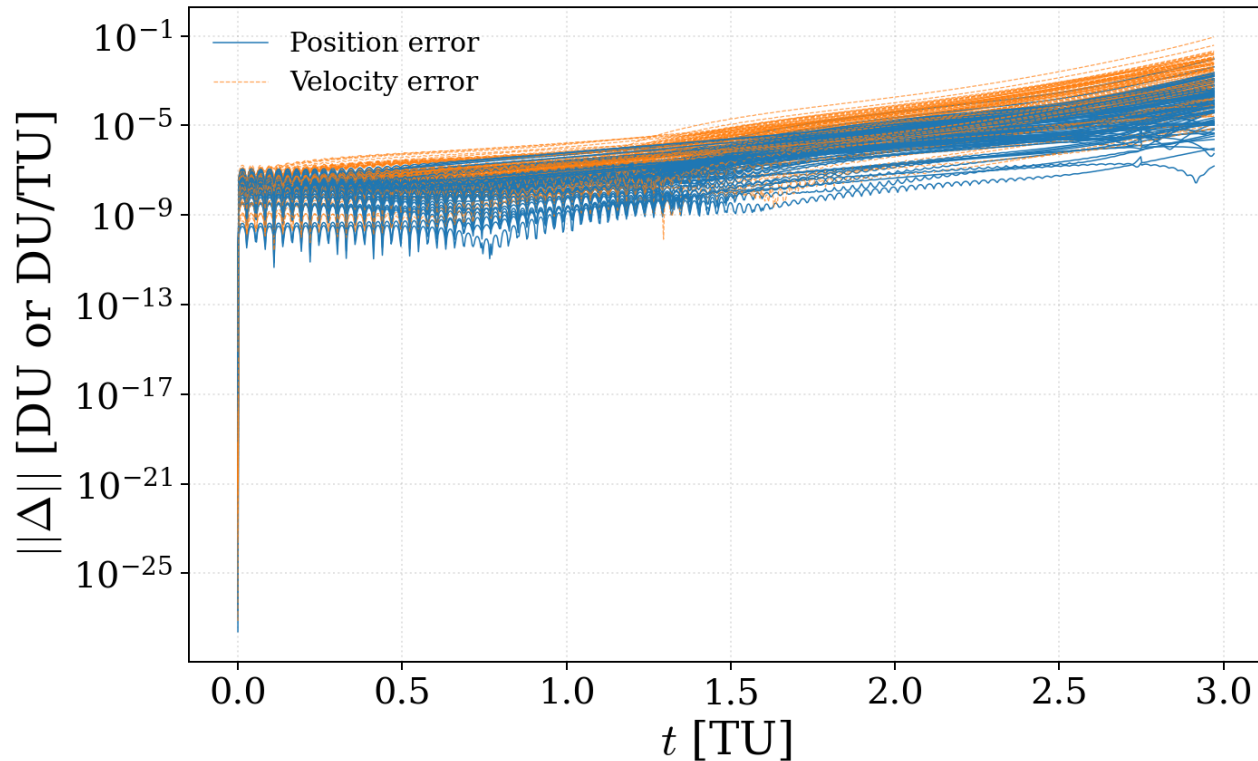
- Example Lyapunov orbit with $C = -3.18$

Floquet multipliers:



Normal Forms around Periodic Orbits

- Propagation in both coordinate systems:



Thank you for your attention!

Jannik Gräbner
jg3607@princeton.edu

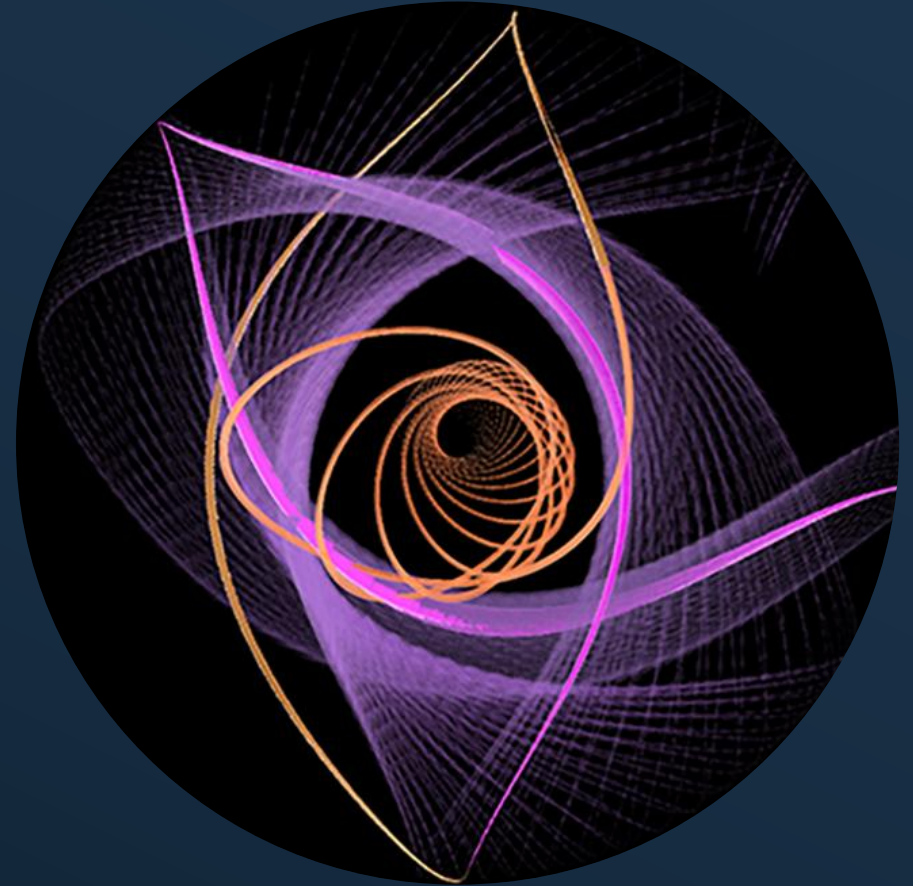
Jack Sydney Schwartz
js1765@princeton.edu

Claudio Vestini
claudio.vestini@keble.ox.ac.uk

Beeson Group



Group Website



Bibliography

- [1] Ralph Abraham and Jerrold E. Marsden. *Foundations of Mechanics*. Addison-Wesley, 2 edition, 1987.
- [2] Roger R. Bate, Donald D. Mueller, and Jerry E. White. *Fundamentals of Astrodynamics*. Dover Publications, New York, 1971.
- [3] Ryne Beeson, Amlan Sinha, Bindu Jagannatha, Devin Bunce, and David Carroll. Dynamically leveraged automated (n) multibody trajectory optimization (dylan). In *AAS/AIAA Space Flight Mechanics Conference*. American Astronautical Society. Charlotte, NC, 2022.
- [4] David Benioff, D. B. Weiss, and Alexander Woo. 3 body problem. Netflix, 2024. TV Series.
- [5] David Brewster. *Memoirs of the Life, Writings, and Discoveries of Sir Isaac Newton*, volume 2. Thomas Constable and Co., Edinburgh, 1855.
- [6] Leonhard Euler. De motu rectilineo trium corporum se mutuo attrahentium. *Novi commentarii academiae scientiarum Petropolitanae*, pages 144–151, 1767.
- [7] Gerard Gómez, Wang Sang Koon, Martin W Lo, Jerrold E Marsden, Josep Masdemont, and Shane D Ross. Invariant manifolds, the spatial three-body problem and petit grand tour of jovian moons. *Celestial Mechanics and Dynamical Astronomy*, 90(1):131–158, 2004.
- [8] Jannik Gräbner. *Towards a Long-Term Average Debris Forecasting Framework for the Cislunar Domain*. Course project, Princeton University, Princeton, NJ, December 2025.
- [9] Naoki Hiraiwa, Mai Bando, Yuzuru Sato, and Shinji Hokamoto. Design of low-energy transfers in cislunar space using sequences of lobe dynamics, 2026. Submitted to *Acta Astronautica*.

Bibliography

- [10] Carl Gustav Jacob Jacobi. Sur le mouvement d'un point et sur un cas particulier du problème des trois corps. *Comptes Rendus de l'Académie des Sciences de Paris*, 3:59–61, 1836.
- [11] Kaitlyn Johnson. Fly Me to the Moon: Worldwide Cislunar and Lunar Missions. Report, Center for Strategic and International Studies (CSIS) Aerospace Security Project, February 2022.
- [12] Àngel Jorba. A methodology for the numerical computation of normal forms, centre manifolds and first integrals of hamiltonian systems. *Experimental Mathematics*, 8(2):155–195, 1999.
- [13] Àngel Jorba and Jordi Villanueva. Numerical computation of normal forms around some periodic orbits of the restricted three-body problem. *Physica D: Nonlinear Phenomena*, 114(3–4):197–229, 1998.
- [14] Wang Sang Koon, Martin W Lo, Jerrold E Marsden, and Shane D Ross. Invariant manifolds, the spatial three-body problem and petit grand tour of jovian moons. *AAS/AIAA Astrodynamics Specialist Conference*, 106:2000–312, 2000.
- [15] Wang Sang Koon, Martin W. Lo, Jerrold E. Marsden, and Shane D. Ross. *Dynamical Systems, the Three-Body Problem and Space Mission Design*. Marsden Books, 2022.
- [16] Joseph-Louis Lagrange. Essai sur le problème des trois corps. *Prix de l'Académie Royale des Sciences de Paris*, 9, 1772.
- [17] Tullio Levi-Civita. Traiettorie singolari ed urti nel problema ristretto dei tre corpi. *Annali di Matematica Pura ed Applicata (1898-1922)*, 9(1):1–32, 1903.
- [18] Isaac Newton. *Philosophiæ Naturalis Principia Mathematica*. Jussu Societatis Regiae ac Typis Josephi Streater, London, 1687.

Bibliography

- [19] Henri Poincaré. Sur le problème des trois corps et les équations de la dynamique. *Acta Mathematica*, 13:1–270, 1890.
- [20] Henri Poincaré. *Les Méthodes Nouvelles de la Mécanique Céleste*, volume 1. Gauthiers-Villars, 1892.
- [21] Henri Poincaré. *Les Méthodes Nouvelles de la Mécanique Céleste*, volume 2. Gauthiers-Villars, 1893.
- [22] Amlan Sinha. *Towards Robust Low-Thrust Trajectory Design Under Missed Thrust Events*. Ph.D. dissertation, Princeton University, Princeton, NJ, May 2026.
- [23] S. Alan Stern. The New Horizons Pluto Kuiper belt mission: An overview with historical context. *Space Science Reviews*, 140(1-4):3–21, 2008.